

4

Inner Product Spaces

SECTION 4.1 Introduction to Inner Product Spaces

By the end of this section you will be able to

- understand what is meant by an inner product space
- prove some properties of inner product spaces
- define and prove properties of the norm of a vector

In chapter 2, we introduced the idea of a dot product for a Euclidean n -space. In this chapter, we extend the concept of a dot product to general vector spaces. The general operation that will replace the dot product in general vector spaces is called an *inner product*. In fact, the dot product was an example of a specific operation more commonly referred to as an inner product, in a Euclidean n -space.

General vector spaces such as the set of polynomials, matrices and continuous functions are fundamentally identical in structure to the Euclidean n -space. Because of this, all of the proofs we introduced for the dot product in Euclidean n -space hold for *any* inner product space. An inner product space is a vector space with additional structure which caters for distance, angle and projection of vectors. We need inner products to define these three notions for general vector spaces.

So far we have only looked at the fundamental linear algebra operations of scalar multiplication and vector addition. We have not mentioned anything about an inner product that involves multiplication of vectors with each other in the general vector space, or inner space.

4.1.1 Definition of inner product

 How did we define the inner or dot product in chapter 2?

Let $\mathbf{u} = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$ be vectors in $\mathbb{R}^n$ then the inner product of $\mathbf{u}$ and $\mathbf{v}$ denoted by $\mathbf{u} \cdot \mathbf{v}$ is

$$(2.4) \quad \mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = u_1 v_1 + u_2 v_2 + u_3 v_3 + \cdots + u_n v_n$$

Remember, the answer was a scalar *not* a vector. This inner product was named the dot product (also called the scalar product) in $\mathbb{R}^n$. This is the usual (or standard) inner product in $\mathbb{R}^n$ but there are many other types of inner products in $\mathbb{R}^n$.

For the general vector space, the inner product is denoted by $\langle \mathbf{u}, \mathbf{v} \rangle$ rather than $\mathbf{u} \cdot \mathbf{v}$. For the general vector space, the definition of inner product is based on Proposition (2.6) of chapter 2 and is given by:

Definition (4.1). An inner product on a real vector space V is an operation which assigns to each pair of vectors, $\mathbf{u}$ and $\mathbf{v}$, a *unique* real number $\langle \mathbf{u}, \mathbf{v} \rangle$ which satisfies the following axioms for *all* vectors $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ in V and *all* scalars k .

- (i) $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$ [commutative law]
- (ii) $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$ [distributive law]
- (iii) $\langle k\mathbf{u}, \mathbf{v} \rangle = k\langle \mathbf{u}, \mathbf{v} \rangle$ [taking out the scalar k]
- (iv) $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$ and we have $\langle \mathbf{u}, \mathbf{u} \rangle = 0 \Leftrightarrow \mathbf{u} = \mathbf{O}$ [Means the inner product between the same vectors is zero or positive.]

A real vector space which satisfies these axioms is called a **real inner product space**. Note that evaluating $\langle , \rangle$ gives a real number (scalar) *not* a vector. Next we give some examples of inner product spaces.

4.1.2 Examples of inner products

Example 4.1

Show that the Euclidean n space, $\mathbb{R}^n$, with the dot product as defined in (2.4) on the previous page is indeed an example of an inner product space.

Solution

See Proposition (2.6) of chapter 2.

The dot product is just one example of an inner product. In fact, there are an infinite number of ways we can define an inner product. The next example demonstrates another inner product on $\mathbb{R}^2$ defined by a matrix sandwiched between two vectors.

Example 4.2

Let the Euclidean 2 space, $\mathbb{R}^2$, be a vector space with

$$\langle \mathbf{u}, \mathbf{w} \rangle = \mathbf{u}^T \mathbf{A} \mathbf{w} \text{ where } \mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$$

Show that this is an inner product for $\mathbb{R}^2$. [Similar to the dot product which is $\mathbf{u}^T \mathbf{w}$, but this time we are required to multiply the vector $\mathbf{w}$ by the matrix $\mathbf{A}$ and then carry out the dot product.]

Solution

Exercises 4.1. [This example works because $\mathbf{A}$ is a symmetric matrix, that is $\mathbf{A}^T = \mathbf{A}$.]

Example 4.3

Let P_2 be the vector space of polynomials of degree 2 or less. Let

$$\mathbf{p} = c_0 + c_1x + c_2x^2 \text{ and } \mathbf{q} = d_0 + d_1x + d_2x^2$$

be polynomials in P_2 . Show that the multiplication of coefficients:

$$\langle \mathbf{p}, \mathbf{q} \rangle = c_0d_0 + c_1d_1 + c_2d_2$$

defines an inner product on P_2 .

[For example, if $\mathbf{p} = 2 + 3x + 5x^2$ and $\mathbf{q} = 4 - x + 7x^2$ then

$$\langle \mathbf{p}, \mathbf{q} \rangle = (2 \times 4) + (3 \times (-1)) + (5 \times 7) = 40]$$

Solution

How do we show $\langle \mathbf{p}, \mathbf{q} \rangle = c_0d_0 + c_1d_1 + c_2d_2$ is an inner product on P_2 ?

Checking *all* four axioms of Definition (4.1).

(i) We need to check $\langle \mathbf{p}, \mathbf{q} \rangle = \langle \mathbf{q}, \mathbf{p} \rangle$ holds for the multiplication of coefficients:

$$\begin{aligned} \langle \mathbf{p}, \mathbf{q} \rangle &= c_0d_0 + c_1d_1 + c_2d_2 \\ &= d_0c_0 + d_1c_1 + d_2c_2 \quad \left[\begin{array}{l} \text{remember, the order of multiplication} \\ \text{does not matter} \end{array} \right] \\ &= \langle \mathbf{q}, \mathbf{p} \rangle \end{aligned}$$

(ii) We need to check $\langle \mathbf{p} + \mathbf{q}, \mathbf{r} \rangle = \langle \mathbf{p}, \mathbf{r} \rangle + \langle \mathbf{q}, \mathbf{r} \rangle$:

Adding the two quadratics $\mathbf{p} = c_0 + c_1x + c_2x^2$ and $\mathbf{q} = d_0 + d_1x + d_2x^2$ gives:

$$\mathbf{p} + \mathbf{q} = (c_0 + d_0) + (c_1 + d_1)x + (c_2 + d_2)x^2$$

Let $\mathbf{r} = e_0 + e_1x + e_2x^2$. Finding the inner product of $\mathbf{p} + \mathbf{q}$ and $\mathbf{r}$:

$$\begin{aligned} \langle \mathbf{p} + \mathbf{q}, \mathbf{r} \rangle &= \langle (c_0 + d_0) + (c_1 + d_1)x + (c_2 + d_2)x^2, e_0 + e_1x + e_2x^2 \rangle \\ &= (c_0 + d_0)e_0 + (c_1 + d_1)e_1 + (c_2 + d_2)e_2 \quad [\text{multiplying coefficients}] \\ &= c_0e_0 + d_0e_0 + c_1e_1 + d_1e_1 + c_2e_2 + d_2e_2 \end{aligned}$$

Evaluating the addition of the other inner products:

$$\begin{aligned} \langle \mathbf{p}, \mathbf{r} \rangle + \langle \mathbf{q}, \mathbf{r} \rangle &= \underbrace{c_0e_0 + c_1e_1 + c_2e_2}_{=\langle \mathbf{p}, \mathbf{r} \rangle} + \underbrace{d_0e_0 + d_1e_1 + d_2e_2}_{=\langle \mathbf{q}, \mathbf{r} \rangle} \\ &= c_0e_0 + d_0e_0 + c_1e_1 + d_1e_1 + c_2e_2 + d_2e_2 \quad [\text{rearranging}] \\ &= \langle \mathbf{p} + \mathbf{q}, \mathbf{r} \rangle \quad [\text{from above}] \end{aligned}$$

Hence part (ii) is satisfied — $\langle \mathbf{p} + \mathbf{q}, \mathbf{r} \rangle = \langle \mathbf{p}, \mathbf{r} \rangle + \langle \mathbf{q}, \mathbf{r} \rangle$.

(continued...)

(iii) We need to check $\langle k\mathbf{p}, \mathbf{q} \rangle = k\langle \mathbf{p}, \mathbf{q} \rangle$ [taking out the scalar k]:

$$\begin{aligned}
 \langle k\mathbf{p}, \mathbf{q} \rangle &= \left\langle k(c_0 + c_1x + c_2x^2), d_0 + d_1x + d_2x^2 \right\rangle \\
 &= \left\langle kc_0 + kc_1x + kc_2x^2, d_0 + d_1x + d_2x^2 \right\rangle \quad [\text{opening brackets}] \\
 &= kc_0d_0 + kc_1d_1 + kc_2d_2 \quad [\text{multiplying coefficients}] \\
 &= k(c_0d_0 + c_1d_1 + c_2d_2) \quad [\text{taking out a common factor}] \\
 &= k\langle \mathbf{p}, \mathbf{q} \rangle
 \end{aligned}$$

Hence part (iii) is satisfied.

(iv) We need to show $\langle \mathbf{p}, \mathbf{p} \rangle \geq 0$ and that we have $\langle \mathbf{p}, \mathbf{p} \rangle = 0 \Leftrightarrow \mathbf{p} = \mathbf{O}$:

$$\begin{aligned}
 \langle \mathbf{p}, \mathbf{p} \rangle &= \left\langle c_0 + c_1x + c_2x^2, c_0 + c_1x + c_2x^2 \right\rangle \\
 &= c_0c_0 + c_1c_1 + c_2c_2 = (c_0)^2 + (c_1)^2 + (c_2)^2 \geq 0
 \end{aligned}$$

Also

$$\langle \mathbf{p}, \mathbf{p} \rangle = (c_0)^2 + (c_1)^2 + (c_2)^2 = 0 \Leftrightarrow c_0 = c_1 = c_2 = 0$$

If $c_0 = c_1 = c_2 = 0$ then $\mathbf{p} = \mathbf{O}$.

All four axioms are satisfied, therefore multiplication of coefficients:

$$\langle \mathbf{p}, \mathbf{q} \rangle = c_0d_0 + c_1d_1 + c_2d_2$$

forms an inner product on P_2 .

In the above example, we defined the inner product as multiplication of each coefficient of the quadratic polynomial:

$$\langle \mathbf{p}, \mathbf{q} \rangle = c_0d_0 + c_1d_1 + c_2d_2$$

However, if we define this as the *addition* of the coefficients:

$$\langle \mathbf{p}, \mathbf{q} \rangle = (c_0 + d_0) + (c_1 + d_1) + (c_2 + d_2)$$

then $\langle \mathbf{p}, \mathbf{q} \rangle$ is *not* an inner product.



Why not?

Because axiom (iv) of Definition (4.1) fails, that is $\langle \mathbf{p}, \mathbf{p} \rangle \geq 0$ is false.

For example, if $\mathbf{p} = -1 - x - x^2$ then

$$\begin{aligned}
 \langle \mathbf{p}, \mathbf{p} \rangle &= \langle -1 - x - x^2, -1 - x - x^2 \rangle \\
 &= (-1 - 1) + (-1 - 1) + (-1 - 1) = -3 < 0
 \end{aligned}$$

Hence $\langle \mathbf{p}, \mathbf{p} \rangle \not\leq 0$ [not greater than or equal to zero].

Next we define an inner product on the vector space of matrices, M_{22} . For this we need to define what is meant by the trace of a matrix:

The **trace** of a matrix is the sum of its leading diagonal elements, that is

$$\text{trace} \begin{pmatrix} \boxed{a} & b \\ c & \boxed{d} \end{pmatrix} = a + d$$

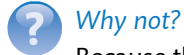
One example of an inner product on the vector space of matrices is the following:

Let M_{22} be the vector space of 2 by 2 matrices and inner product on M_{22} be defined by

$$\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{B}^T \mathbf{A})$$

where tr is the trace. This is an inner product on M_{22} – you may like to check all 4 axioms.

If we define the inner product as our normal matrix multiplication $\langle \mathbf{A}, \mathbf{B} \rangle = \mathbf{AB}$ on M_{22} then this is *not* an inner product.



Why not?

Because the commutative law does not hold, that is

$$\langle \mathbf{A}, \mathbf{B} \rangle = \mathbf{AB} \neq \mathbf{BA} = \langle \mathbf{B}, \mathbf{A} \rangle \text{ [not equal]}$$

There are many other examples of inner product spaces which you are asked to verify in Exercises 4.1. Next we move onto proving properties of general inner products.

4.1.3 Properties of inner products

Proposition (4.2). Let $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ be vectors in a real inner product space V and k be any real scalar. We have the following properties of inner products:

- (i) $\langle \mathbf{u}, \mathbf{0} \rangle = \langle \mathbf{0}, \mathbf{v} \rangle = 0$
- (ii) $\langle \mathbf{u}, k\mathbf{v} \rangle = k\langle \mathbf{u}, \mathbf{v} \rangle$
- (iii) $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$



How do we prove these properties?

We use the axioms of inner products stated in Definition (4.1).

Proof of (i).

We can write the zero vector as $0(\mathbf{0})$ because $0(\mathbf{0}) = \mathbf{0}$. Using the axioms of definition (4.1) we have

$$\begin{aligned} \langle \mathbf{u}, \mathbf{0} \rangle &= \langle \mathbf{u}, 0(\mathbf{0}) \rangle \\ &= \langle 0(\mathbf{0}), \mathbf{u} \rangle && [\text{by part (i) of (4.1) which is } \langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle] \\ &= 0\langle \mathbf{0}, \mathbf{u} \rangle && [\text{by part (iii) of (4.1) which is } \langle k\mathbf{u}, \mathbf{v} \rangle = k\langle \mathbf{u}, \mathbf{v} \rangle] \\ &= 0 \end{aligned}$$

Similarly $\langle \mathbf{0}, \mathbf{v} \rangle = 0$.



Proof of (ii).

The inner product is commutative, $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$, which means we can switch the vectors around. We have

$$\begin{aligned} \langle \mathbf{u}, k\mathbf{v} \rangle &= \langle k\mathbf{v}, \mathbf{u} \rangle && [\text{switching vectors } \langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle] \\ &= k\langle \mathbf{v}, \mathbf{u} \rangle && [\text{by part (iii) of (4.1) which is } \langle k\mathbf{u}, \mathbf{v} \rangle = k\langle \mathbf{u}, \mathbf{v} \rangle] \\ &= k\langle \mathbf{u}, \mathbf{v} \rangle && [\text{switching vectors } \langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle] \end{aligned}$$

■

Proof of (iii).

We have

$$\begin{aligned} \langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle &= \langle \mathbf{v} + \mathbf{w}, \mathbf{u} \rangle && [\text{switching vectors } \langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle] \\ &= \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{w}, \mathbf{u} \rangle && \left[\begin{array}{l} \text{by part (ii) of (4.1) which is} \\ \langle \mathbf{v} + \mathbf{w}, \mathbf{u} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{w}, \mathbf{u} \rangle \end{array} \right] \\ &= \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle && [\text{switching vectors } \langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle] \end{aligned}$$

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4.1.4 The norm or length of a vector

From chapter 2 we know that we can find the length (norm) $\|\mathbf{u}\|$ of a vector $\mathbf{u}$.



Do you remember how the (Euclidean) norm was defined?

The norm or length of a vector $\mathbf{u}$ in $\mathbb{R}^n$ was defined by Pythagoras' Theorem: $\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}}$.

The norm is defined in the same manner for the general vector space V . Let $\mathbf{u}$ be a vector in V then the norm denoted by $\|\mathbf{u}\|$ is defined as

$$(4.3) \quad \|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle} \text{ [positive root]}$$

Note that for the general vector space we *cannot* use the definition for the dot product because that is only defined for Euclidean space, $\mathbb{R}^n$, and in this chapter we are examining inner products on general vector spaces. The norm of a vector $\mathbf{u}$ is a real number which gives the size of the vector $\mathbf{u}$.

Generally to find the norm $\|\mathbf{u}\|$ we find $\|\mathbf{u}\|^2 = \langle \mathbf{u}, \mathbf{u} \rangle$ and then take the square root of the result.



How do we find the norm of a function or a matrix?

The norm of a matrix $\mathbf{A}$ in the vector space M_{mn} is given by the above (4.3):

$$\|\mathbf{A}\| = \sqrt{\langle \mathbf{A}, \mathbf{A} \rangle} \text{ where } \langle \mathbf{A}, \mathbf{A} \rangle \text{ is the inner product of } \mathbf{A} \text{ and } \mathbf{A}.$$

Norm measures the magnitude of things. For example, if take our earlier inner product defined for matrices, $\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{B}^T \mathbf{A})$, then we can evaluate the magnitude of $\|\mathbf{A}\|$ and $\|\mathbf{B}\|$ by calculating the norms.

Let $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 10 & 20 \\ 30 & 40 \end{pmatrix}$, find $\|\mathbf{A}\|$ and $\|\mathbf{B}\|$.

We have

$$\begin{aligned}\|\mathbf{A}\|^2 &= \langle \mathbf{A}, \mathbf{A} \rangle = \text{tr} \left[\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}^T \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \right] \\ &= \text{tr} \left[\begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \right] = \text{tr} \left[\begin{pmatrix} 10 & * \\ * & 20 \end{pmatrix} \right] = 10 + 20 = 30\end{aligned}$$

Hence $\|\mathbf{A}\| = \sqrt{30}$. Since the inner product is the trace of the matrix, we don't need to worry what the entries on the other diagonal are; that is why we have placed * in those positions.

Similarly

$$\|\mathbf{B}\| = \sqrt{3000} = 10\sqrt{30} = 10\|\mathbf{A}\| \quad [\text{because } \|\mathbf{A}\| = \sqrt{30}]$$

Notice that the entries in matrix $\mathbf{B}$ are 10 times the entries in matrix $\mathbf{A}$. The norm of matrix $\mathbf{B}$ is 10 times the norm of matrix $\mathbf{A}$.



Why are we interested in the magnitude of a matrix?

Remember, matrices are functions applied to vectors such as $\mathbf{x}$, say. We generally want to compare the size of vectors $\mathbf{Ax}$ with $\mathbf{Bx}$, which is given by their norms $\|\mathbf{Ax}\| = \|\mathbf{A}\| \|\mathbf{x}\|$ and $\|\mathbf{Bx}\| = \|\mathbf{B}\| \|\mathbf{x}\|$. In the above example, the vector $\mathbf{Bx}$ is 10 times larger in magnitude than vector $\mathbf{Ax}$.

In the following example we apply the norm to vector spaces of continuous functions.

Example 4.4

Let $C[0, 1]$ be the vector space of continuous functions on the closed interval $[0, 1]$. Let $f(x) = x$ and $g(x) = x^2 - 1$ be functions in $C[0, 1]$ and their inner product be given by

$$\langle \mathbf{f}, \mathbf{g} \rangle = \int_0^1 f(x)g(x) dx$$

This inner product requires us to multiply two vectors and then sum them over $[0, 1]$. [Similar to the dot product but in place of a finite sum we have an integral because we are dealing with continuous functions.] Determine:

- (i) $\langle \mathbf{f}, \mathbf{g} \rangle$ (ii) $\|\mathbf{f}\|$ (iii) $\|\mathbf{g}\|$ (iv) $\|\mathbf{f} - \mathbf{g}\|$ (v) $\|\mathbf{f} - \mathbf{h}\|$ where $h(x) = x^3$

Solution

- (i) The inner product $\langle \mathbf{f}, \mathbf{g} \rangle = \int_0^1 f(x)g(x) dx$ with $f(x) = x$ and $g(x) = x^2 - 1$ gives

(continued...)

$$\begin{aligned}
\langle \mathbf{f}, \mathbf{g} \rangle &= \int_0^1 f(x) g(x) dx \\
&= \int_0^1 x(x^2 - 1) dx \quad \left[\text{substituting } f(x) = x \text{ and } g(x) = x^2 - 1 \right] \\
&= \int_0^1 (x^3 - x) dx = \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_0^1 = \left[\frac{1}{4} - \frac{1}{2} \right] = -\frac{1}{4}
\end{aligned}$$

(ii) As stated above, to find $\|\mathbf{f}\|$ we determine $\|\mathbf{f}\|^2 = \langle \mathbf{f}, \mathbf{f} \rangle$ and then take the square root of the result:

$$\begin{aligned}
\|\mathbf{f}\|^2 &= \langle \mathbf{f}, \mathbf{f} \rangle \\
&= \int_0^1 f(x) f(x) dx \quad \stackrel{\text{substituting } f(x)=x}{=} \int_0^1 x x dx = \int_0^1 x^2 dx = \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{3} \quad \left[\text{because } xx = x^2 \right]
\end{aligned}$$

What is $\|\mathbf{f}\|$ equal to?

Square root of $\|\mathbf{f}\|^2 = \frac{1}{3}$ which is $\|\mathbf{f}\| = \frac{1}{\sqrt{3}}$.

What does this $\|\mathbf{f}\| = \frac{1}{\sqrt{3}}$ signify?

The norm $\|\mathbf{f}\| = \left(\int_0^1 [f(x)]^2 dx \right)^{1/2}$ involves integrating (summing) the function squared, and then taking the square root which measures an average value of the function, called the *root mean square – rms*.

$\|\mathbf{f}\| = 1/\sqrt{3} = 0.577$ (3dp) is the rms value of the function $f(x) = x$ between 0 and 1.

(iii) Similarly we have

$$\begin{aligned}
\|\mathbf{g}\|^2 &= \langle \mathbf{g}, \mathbf{g} \rangle = \int_0^1 g(x) g(x) dx \\
&= \int_0^1 (x^2 - 1)(x^2 - 1) dx \quad \left[\text{substituting } g(x) = x^2 - 1 \right] \\
&= \int_0^1 (x^4 - 2x^2 + 1) dx = \left[\frac{x^5}{5} - 2\frac{x^3}{3} + x \right]_0^1 = \frac{1}{5} - \frac{2}{3} + 1 = \frac{8}{15} \quad \left[\text{integrating} \right]
\end{aligned}$$

What is $\|\mathbf{g}\|$ equal to?

We need to take the square root of $\frac{8}{15}$ to find $\|\mathbf{g}\|$. Hence $\|\mathbf{g}\| = \sqrt{\frac{8}{15}}$ signifies the rms value, which is an average value of the function $g(x) = x^2 - 1$ between 0 and 1. Note that this $\|\mathbf{g}\|$ is always a positive value, despite $g(x) = x^2 - 1$ being negative between 0 and 1, because we have squared the function and then taken the positive square root.

(iv) Similarly to find $\|\mathbf{f} - \mathbf{g}\|$ we first determine $\|\mathbf{f} - \mathbf{g}\|^2$ and then take the square root:

$$\begin{aligned}
 \|\mathbf{f} - \mathbf{g}\|^2 &= \langle \mathbf{f} - \mathbf{g}, \mathbf{f} - \mathbf{g} \rangle \\
 &= \int_0^1 [f(x) - g(x)] [f(x) - g(x)] dx \\
 &= \int_0^1 [x - x^2 - 1] [x - x^2 - 1] dx && \text{[substituting]} \\
 &= \int_0^1 (x^2 - x^3 - x - x^3 + x^4 + x^2 - x + x^2 + 1) dx && \text{[expanding]} \\
 &= \int_0^1 (x^4 - 2x^3 + 3x^2 - 2x + 1) dx && \text{[simplifying]} \\
 &= \left[\frac{x^5}{5} - 2\frac{x^4}{4} + 3\frac{x^3}{3} - 2\frac{x^2}{2} + x \right]_0^1 && \text{[integrating]} \\
 &= \left[\frac{x^5}{5} - \frac{x^4}{2} + x^3 - x^2 + x \right]_0^1 = \left[\frac{1}{5} - \frac{1}{2} + 1 - 1 + 1 \right] = \frac{7}{10}
 \end{aligned}$$

Hence $\|\mathbf{f} - \mathbf{g}\| = \sqrt{7/10}$.

(v) Similarly $\|\mathbf{f} - \mathbf{h}\| = \sqrt{8/105}$.

$\|\mathbf{f} - \mathbf{g}\|$ measures the distance between functions $\mathbf{f}$ and $\mathbf{g}$ and is critical in the study of mathematical and numerical analysis. A measurement of distance between functions tells us how close together these functions are.

The **distance** or **metric** between two vectors $\mathbf{u}$ and $\mathbf{v}$ is denoted by $d(\mathbf{u}, \mathbf{v})$ and is defined as

$$(4.4) \quad d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$$

In the above Example 4.4, the distance between $\mathbf{f}$ and $\mathbf{g}$ was given by $\|\mathbf{f} - \mathbf{g}\| = \sqrt{7/10}$ and the distance between $\mathbf{f}$ and $\mathbf{h}$ was $\|\mathbf{f} - \mathbf{h}\| = \sqrt{8/105}$. Clearly $\mathbf{f}$ is closer to $\mathbf{h}$ than $\mathbf{g}$ because $\sqrt{8/105} = 0.276$ is smaller than $\sqrt{7/10} = 0.837$.

Remember, we have introduced $\mathbf{f}$ and $\mathbf{g}$ as vectors in a vector space of $C[0, 1]$. We can view this as shown in Fig. 4.1.

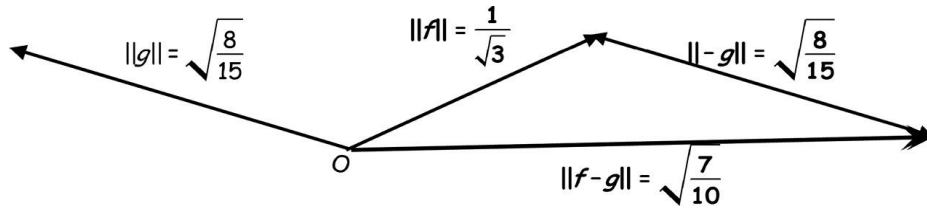


Figure 4.1

We use inner products to measure how close one function is to another. This is important in the study of numerical analysis.



What does $\|\mathbf{f} - \mathbf{g}\| = 0$ mean?

The functions $\mathbf{f}$ and $\mathbf{g}$ are equal because

$$\|\mathbf{f} - \mathbf{g}\| = 0 \Rightarrow \langle \mathbf{f} - \mathbf{g}, \mathbf{f} - \mathbf{g} \rangle = 0 \Rightarrow \mathbf{f} = \mathbf{g}$$

Next we look at a norm on the set of polynomials P_n .

Example 4.5

Determine the norm $\|\cdot\|$ of the quadratic (vector) in P_2 , where the inner product is given by

$$\langle \mathbf{p}, \mathbf{q} \rangle = c_0 d_0 + c_1 d_1 + c_2 d_2 \quad [\text{multiplication of coefficients}]$$

for (i) $\mathbf{p} = x^2 + x + 1$

(ii) $\mathbf{q} = 2x^2 + 2x + 2$

(iii) $\mathbf{r} = c_2 x^2 + c_1 x + c_0$

Solution

We can plot these vectors in 3d space with axes labelled x^2 , x and constant. These axes are *not* perpendicular to each other as we will show in section 4.3.

Since $\mathbf{p} = x^2 + x + 1$, the x^2 , x and constant coefficients for $\mathbf{p}$ are 1, 1 and 1 respectively. Similarly we plot the other vectors (Fig. 4.2).

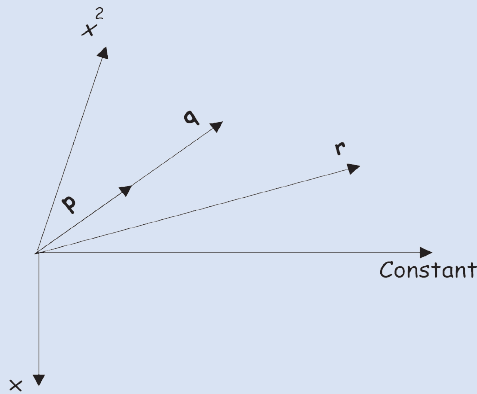


Figure 4.2

We will show in section 4.3 that the constant and x axes are, in fact, perpendicular to each other. However, the x^2 axis is not perpendicular to these.

(i) By definition (4.3) above we have $\|\mathbf{p}\| = \sqrt{\langle \mathbf{p}, \mathbf{p} \rangle}$ so first we find $\|\mathbf{p}\|^2 = \langle \mathbf{p}, \mathbf{p} \rangle$ and then take the square root:

$$\langle \mathbf{p}, \mathbf{p} \rangle = \langle x^2 + x + 1, x^2 + x + 1 \rangle = 1^2 + 1^2 + 1^2 = 3$$

Taking the square root yields $\|\mathbf{p}\| = \sqrt{3}$.

(ii) Similarly we have

$$\langle \mathbf{q}, \mathbf{q} \rangle = \langle 2x^2 + 2x + 2, 2x^2 + 2x + 2 \rangle = 2^2 + 2^2 + 2^2 = 12$$

Hence $\|\mathbf{q}\| = \sqrt{12} = \sqrt{4 \times 3} = 2\sqrt{3}$.

(iii) Again, repeating this process for $\mathbf{r} = c_2 x^2 + c_1 x + c_0$:

$$\langle \mathbf{r}, \mathbf{r} \rangle = c_0 c_0 + c_1 c_1 + c_2 c_2 = (c_0)^2 + (c_1)^2 + (c_2)^2$$

Therefore $\|\mathbf{r}\| = \sqrt{\langle \mathbf{r}, \mathbf{r} \rangle} = \sqrt{(c_0)^2 + (c_1)^2 + (c_2)^2}$.

This $\|\mathbf{r}\|$ gives the size of the quadratic by applying Pythagoras to each of the coefficients (legs), c_0 , c_1 and c_2 .

Note that, for this Example 4.5, we had $\|\mathbf{p}\| = \sqrt{3}$ and $\|\mathbf{q}\| = 2\sqrt{3}$, which means that the quadratic $\mathbf{q} = 2x^2 + 2x + 2$ is twice the size of $\mathbf{p} = x^2 + x + 1$, and note that $\mathbf{q} = 2\mathbf{p}$.

Some norms are *not* defined in terms of the inner product.

Consider the Euclidean space $\mathbb{R}^n$ and let $\mathbf{x} = (x_1 \ x_2 \ \cdots \ x_n)^T$ be a vector in $\mathbb{R}^n$, then the following are important norms in this space:

$$\|\mathbf{x}\|_1 = |x_1| + |x_2| + |x_3| + \cdots + |x_n| \quad [\text{This is called the } \mathbf{one} \text{ norm}]$$

$$\|\mathbf{x}\|_2 = \sqrt{|x_1|^2 + |x_2|^2 + |x_3|^2 + \cdots + |x_n|^2} \quad [\text{This is called the } \mathbf{two} \text{ norm}]$$

$$\|\mathbf{x}\|_\infty = \max(|x_j|) \text{ where } j = 1, 2, \dots, n \quad [\text{This is called the } \mathbf{infinity} \text{ norm}]$$

Note that $\|\mathbf{x}\|_\infty = \max(|x_j|)$ where $j = 1, 2, \dots, n$ means that we select the maximum absolute value out of $x_1, x_2, x_3, \dots$ and x_n . This norm measures the maximum absolute value. For example, if we have the vector $\mathbf{x} = (-1 \ 4 \ -9 \ 7)^T$ in $\mathbb{R}^4$ then

$$\|\mathbf{x}\|_1 = |-1| + |4| + |-9| + |7| = 21$$

$$\|\mathbf{x}\|_2 = \sqrt{|-1|^2 + |4|^2 + |-9|^2 + |7|^2} = \sqrt{147} = 12.12 \text{ (2 dp)}$$

$$\|\mathbf{x}\|_\infty = \max(|-1|, |4|, |-9|, |7|) = \max(1, 4, 9, 7) = 9$$

Which norm should we use?

In a physical situation, you must decide which of these norms suits your problem. For example, suppose we want to measure the noise on a signal, which means that we need a way to measure the variation from the main signal. For the above defined norms, $\|\mathbf{x}\|_\infty$ would measure the largest difference, $\|\mathbf{x}\|_2$ would measure the added average variation and $\|\mathbf{x}\|_1$ would measure the total difference from the main signal. Once you have decided which measure is most appropriate then you try to minimize this.

4.1.5 Properties of the norm of a vector

Next we state certain properties of the norm of a vector.

Proposition (4.5). Let V be an inner product space and $\mathbf{u}$ and $\mathbf{v}$ be vectors in V . If k is any real scalar then we have the following properties of norms:

$$(i) \|\mathbf{u}\| \geq 0 \text{ [non-negative]} \quad (ii) \|\mathbf{u}\| = 0 \Leftrightarrow \mathbf{u} = \mathbf{O} \quad (iii) \|k\mathbf{u}\| = |k| \|\mathbf{u}\|$$

[Note that for a real scalar k we have $\sqrt{k^2} = |k|$ where $|k|$ is the modulus of k .]

How do we prove these results?

We use the definition of the norm given above:

$$(4.3) \quad \|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}$$

Proof of part (i).

By the definition of a norm (4.3) $\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}$ we have

$$\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle} \geq 0$$

Proof of part (ii) which is $\|\mathbf{u}\| = 0 \Leftrightarrow \mathbf{u} = \mathbf{O}$.

Again, by the norm definition (4.3) we have

$$\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle} = 0 \Rightarrow \mathbf{u} = \mathbf{O}$$

and

$$\mathbf{u} = \mathbf{O} \Rightarrow \|\mathbf{u}\| = \sqrt{\langle \mathbf{O}, \mathbf{O} \rangle} = 0$$

Proof of part (iii), which is $\|k\mathbf{u}\| = |k| \|\mathbf{u}\|$.

Applying the norm definition (4.3), we first find $\|k\mathbf{u}\|^2$ and then we take the square root:

$$\begin{aligned} \|k\mathbf{u}\|^2 &= \langle k\mathbf{u}, k\mathbf{u} \rangle \\ &= k \langle \mathbf{u}, k\mathbf{u} \rangle \\ &= kk \langle \mathbf{u}, \mathbf{u} \rangle = k^2 \langle \mathbf{u}, \mathbf{u} \rangle \end{aligned}$$

Taking the square root gives

$$\|k\mathbf{u}\| = \sqrt{k^2 \langle \mathbf{u}, \mathbf{u} \rangle} = \sqrt{k^2} \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle} = |k| \|\mathbf{u}\|$$

This is our required result.



Summary

(4.1). An inner product is a function on a vector space which satisfies the four axioms.

The norm or length of a vector denoted $\|\mathbf{u}\|$, is defined as $\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}$.



EXERCISES 4.1

(Brief solutions at end of book. Full solutions available at <http://www.oup.co.uk/companion/singh>.)

1. Consider the inner product defined on $C[0, 1]$ defined in Example 4.4.

If $f(x) = x$ and $g(x) = x^2$ then determine the following:

- | | | | |
|--|--|---|--|
| (a) $\langle \mathbf{f}, \mathbf{g} \rangle$ | (b) $\langle \mathbf{g}, \mathbf{f} \rangle$ | (c) $\langle 3\mathbf{f}, \mathbf{g} \rangle$ | (d) $\langle \mathbf{f}, \mathbf{f} \rangle$ |
| (e) $\ \mathbf{f}\ $ | (f) $\langle \mathbf{g}, \mathbf{g} \rangle$ | (g) $\ \mathbf{g}\ $ | |

2. Let $C[-1, 1]$ be the vector space of all the continuous functions on the interval $[-1, 1]$. The following is an inner product on $C[-1, 1]$:

$$\langle \mathbf{f}, \mathbf{g} \rangle = \int_{-1}^1 f(x)g(x) dx$$

If $f(x) = x$ and $g(x) = x^3$ then determine the following:

- (a) $\langle \mathbf{f}, \mathbf{g} \rangle$ (b) $\langle \mathbf{g}, \mathbf{f} \rangle$ (c) $\langle 3\mathbf{f}, \mathbf{g} \rangle$ (d) $\langle \mathbf{f}, \mathbf{f} \rangle$
 (e) $\|\mathbf{f}\|$ (f) $\langle \mathbf{g}, \mathbf{g} \rangle$ (g) $\|\mathbf{g}\|$

3. Let $P_2(x)$ be the vector space of polynomials of degree 2 or less. Consider the inner product on this space given in Example 4.3.

If $\mathbf{p} = 2 - 3x + 5x^2$ and $\mathbf{q} = 7 + 5x - 4x^2$ then determine the following:

- (a) $\langle \mathbf{p}, \mathbf{q} \rangle$ (b) $\langle \mathbf{q}, \mathbf{p} \rangle$ (c) $\langle \mathbf{p}, -3\mathbf{q} \rangle$ (d) $\langle \mathbf{p}, \mathbf{p} \rangle$
 (e) $\|\mathbf{p}\|$ (f) $\langle \mathbf{q}, \mathbf{q} \rangle$ (g) $\|\mathbf{q}\|$

4. Show that $\langle \mathbf{u}, \mathbf{w} \rangle = \mathbf{u}^T \mathbf{A} \mathbf{w}$, as defined in Example 4.2, is indeed an inner product.
 5. Let M_{22} be the vector space of 2 by 2 matrices and inner product on M_{22} be defined by

$$\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{B}^T \mathbf{A})$$

where tr is the trace. For $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} -1 & 1 \\ 2 & 5 \end{pmatrix}$ determine

- (a) $\langle \mathbf{A}, \mathbf{B} \rangle$ (b) $\langle 5\mathbf{A}, \mathbf{B} \rangle$ (c) $\langle -\mathbf{A}, -\mathbf{B} \rangle$ (d) $\|\mathbf{A}\|$ (e) $\|\mathbf{B}\|$
 (f) $\langle \mathbf{A}, \mathbf{C} \rangle$ (g) $\langle \mathbf{B}, \mathbf{C} \rangle$ (h) $\langle \mathbf{A} + \mathbf{B}, \mathbf{C} \rangle$ (i) $\langle \mathbf{A}, \mathbf{C} + \mathbf{B} \rangle$

6. Let V be an inner product space. Prove the following results in the inner product space:

- (a) $\langle \mathbf{u}, \mathbf{v} - \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{w} \rangle$
 (b) $\langle \mathbf{u} - \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle - \langle \mathbf{v}, \mathbf{w} \rangle$
 (c) $\langle k_1 \mathbf{u}, k_2 \mathbf{v} \rangle = k_1 k_2 \langle \mathbf{u}, \mathbf{v} \rangle$
 (d) $\langle k_1 \mathbf{u} + k_2 \mathbf{v}, k_3 \mathbf{w} + k_4 \mathbf{x} \rangle = k_1 k_3 \langle \mathbf{u}, \mathbf{w} \rangle + k_1 k_4 \langle \mathbf{u}, \mathbf{x} \rangle + k_2 k_3 \langle \mathbf{v}, \mathbf{w} \rangle + k_2 k_4 \langle \mathbf{v}, \mathbf{x} \rangle$
 (e) $\langle k_1 \mathbf{u}_1 + k_2 \mathbf{u}_2 + \cdots + k_n \mathbf{u}_n, \mathbf{w} \rangle = k_1 \langle \mathbf{u}_1, \mathbf{w} \rangle + k_2 \langle \mathbf{u}_2, \mathbf{w} \rangle + \cdots + k_n \langle \mathbf{u}_n, \mathbf{w} \rangle$

7. Let $C[0, 1]$ be the vector space of all the continuous functions on the interval $[0, 1]$. Consider the normal inner product of this space given in Example 4.4.

If $f(x) = x + 1$, $g(x) = x^2$ and $h(x) = x - 1$ then determine the following:

- (a) $\langle \mathbf{f}, \mathbf{g} \rangle$ (b) $\langle \mathbf{g}, \mathbf{h} \rangle$ (c) $\langle \mathbf{f}, \mathbf{h} \rangle$ (d) $\langle \mathbf{f} + \mathbf{g}, \mathbf{h} \rangle$
 (e) $\langle \mathbf{f}, \mathbf{h} + \mathbf{g} \rangle$ (f) $\langle \mathbf{f} + 3\mathbf{g}, \mathbf{h} \rangle$ (g) $\langle \mathbf{f} - 3\mathbf{g}, \mathbf{h} \rangle$ (h) $\langle \mathbf{f} - \mathbf{g}, \mathbf{h} \rangle$
 (i) $\langle 2\mathbf{f} + 5\mathbf{g}, 6\mathbf{h} \rangle$ (j) $\langle -10\mathbf{f}, 2\mathbf{h} + 5\mathbf{g} \rangle$

8. Let $C[0, 1]$ be the vector space of all the continuous functions on the interval $[0, 1]$. Explain why the following is not an inner product on $C[0, 1]$:

$$\langle \mathbf{f}, \mathbf{g} \rangle = \int_0^1 [f(x) - g(x)] dx$$

9. Show that $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{A} \mathbf{v}$, where $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ is *not* an inner product on $\mathbb{R}^2$.
10. Show that $\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + u_2 v_2 + u_3 v_3 - u_4 v_4$, where $\mathbf{u} = (u_1 \ u_2 \ u_3 \ u_4)^T$ and $\mathbf{v} = (v_1 \ v_2 \ v_3 \ v_4)^T$ are vectors in $\mathbb{R}^4$, is *not* an inner product on $\mathbb{R}^4$.
11. Let $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ be vectors in an inner product space V . If

$$\langle \mathbf{u}, \mathbf{v} \rangle = -1, \langle \mathbf{u}, \mathbf{w} \rangle = 5, \langle \mathbf{v}, \mathbf{w} \rangle = 3, \|\mathbf{v}\| = 2, \|\mathbf{u}\| = 4 \text{ and } \|\mathbf{w}\| = 7,$$

then determine the following:

$$(a) \langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle \quad (b) \langle 2\mathbf{u} + 3\mathbf{v}, 5\mathbf{v} - 2\mathbf{w} \rangle \quad (c) \|\mathbf{u} - \mathbf{v}\| \quad (d) \|\mathbf{u} - \mathbf{v} - \mathbf{w}\|$$

12. In (4.4) we defined the distance function as

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$$

where $\mathbf{u}$ and $\mathbf{v}$ are vectors in V . Prove the following properties of this distance function:

$$(a) d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u}) \quad (b) d(\mathbf{u}, \mathbf{v}) \geq 0 \quad (c) d(\mathbf{u}, \mathbf{v}) = 0 \Leftrightarrow \mathbf{u} = \mathbf{v} \\ (d) d(k\mathbf{u}, k\mathbf{v}) = |k| d(\mathbf{u}, \mathbf{v})$$

SECTION 4.2 Inequalities and Orthogonality

By the end of this section you will be able to

- state and prove Cauchy-Schwarz and Minkowski inequalities
- understand what is meant by orthogonal, normalized and orthonormal vectors

A common application of the inner product is to establish when two or more vectors are orthogonal, or perpendicular to each other.



Why is orthogonality important?

For general vector spaces such as polynomials, matrices and functions, a simple basis is one where the set of vectors are mutually orthogonal (perpendicular) and have unit length. We will discuss the concept of a basis in the next section.

First, we do a numerical example involving the norm and the inner product in $\mathbb{R}^2$. In this example, the inner product is the usual dot product for $\mathbb{R}^2$.

Example 4.6

Let $\mathbf{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ be in $\mathbb{R}^2$. Determine the following with respect to the dot product:

- (i) $|\langle \mathbf{u}, \mathbf{v} \rangle|$ (ii) $\|\mathbf{u}\| \|\mathbf{v}\|$ (iii) $\|\mathbf{u}\| + \|\mathbf{v}\|$ (iv) $\|\mathbf{u} + \mathbf{v}\|$

Solution

(i) We have $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 4 \end{pmatrix} = (1 \times 3) + (2 \times 4) = 11$. Hence $|\langle \mathbf{u}, \mathbf{v} \rangle| = |11| = 11$.

(ii) Remember, the symbol $\|\cdot\|$ means the length of the vector. By Pythagoras we have

$$\|\mathbf{u}\| = \left\| \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\| = \sqrt{1^2 + 2^2} = \sqrt{5} \quad \text{and} \quad \|\mathbf{v}\| = \left\| \begin{pmatrix} 3 \\ 4 \end{pmatrix} \right\| = \sqrt{3^2 + 4^2} = 5$$

Multiplying these we have $\|\mathbf{u}\| \|\mathbf{v}\| = \sqrt{5} \times 5 = 11.1803$ (4dp).

(iii) Adding both the above results in part (ii) we have

$$\|\mathbf{u}\| + \|\mathbf{v}\| = \sqrt{5} + 5 = 7.2361 \text{ (4dp)}$$

(iv) Similarly by using Pythagoras we have

$$\|\mathbf{u} + \mathbf{v}\| = \left\| \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ 4 \end{pmatrix} \right\| = \left\| \begin{pmatrix} 4 \\ 6 \end{pmatrix} \right\| = \sqrt{4^2 + 6^2} = \sqrt{52} = 7.2111 \text{ (4 dp)}$$

We can illustrate these in $\mathbb{R}^2$ (Fig. 4.3):

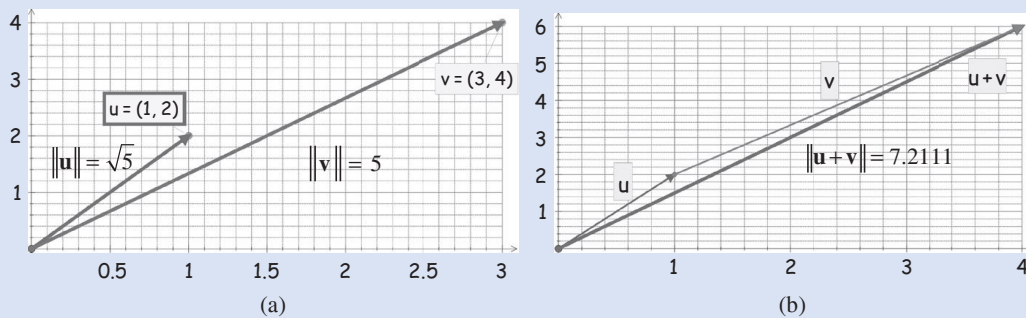


Figure 4.3

4.2.1 Inequalities

In this subsection we prove two inequalities:

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\| \quad \text{Cauchy-Schwarz inequality}$$

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\| \quad \text{Minkowski inequality}$$

First we state and prove the Cauchy-Schwarz inequality. It is an important inequality which is used in many applications and fields of mathematics. It is a slightly more involved proof because we apply one particular result of quadratic equations in order to prove the Cauchy-Schwarz inequality.



Under what conditions is the general quadratic $ax^2 + bx + c \geq 0$ (greater than or equal to zero)?

Well, the graph of such a quadratic lies above (or on) the x axis, as shown in Fig. 4.4:

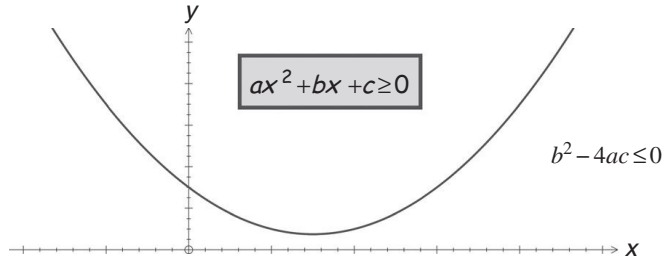


Figure 4.4

For the quadratic, $ax^2 + bx + c$, to be greater than or equal to 0 for all x means that either the graph does not cross the x axis or it touches it at just one point. This is satisfied when

$$b^2 - 4ac \leq 0 \quad (\dagger)$$

This is where the discriminant, $b^2 - 4ac$, is less than or equal to zero. [Remember, the discriminant in this case means that it discriminates between real and complex roots of a quadratic equation.]

The Cauchy–Schwarz inequality for Euclidean space $\mathbb{R}^n$ was covered in section 2.2.3.

It demonstrates a universal inequality between inner products and norms, and was given by:

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

Here we expressed the inequality in terms of the dot product, but we can extend this definition to the general inner product:

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

The Cauchy–Schwarz inequality (4.6). Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in an inner product space, then

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

We can see this inequality for the above Example 4.6:

$$\begin{array}{ccccc} |\langle \mathbf{u}, \mathbf{v} \rangle| & \equiv & 11 & \leq & 11.1803 & \equiv & \|\mathbf{u}\| \|\mathbf{v}\| \\ & \text{by part (i)} & & & & \text{by part (ii)} & \end{array}$$

Note that the Cauchy–Schwarz inequality connects the notion of an inner product with the notion of length.

[Hint: Consider the inner product $\langle k\mathbf{u} + \mathbf{v}, k\mathbf{u} + \mathbf{v} \rangle$ where k is a scalar.]

Proof.



How do we prove this result?

By the hint, expanding the given inner product $\langle k\mathbf{u} + \mathbf{v}, k\mathbf{u} + \mathbf{v} \rangle$ and by using the rules on inner products established in the last section:

$$\begin{aligned}\langle k\mathbf{u} + \mathbf{v}, k\mathbf{u} + \mathbf{v} \rangle &= \langle k\mathbf{u}, k\mathbf{u} \rangle + \langle k\mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, k\mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle \\ &= k^2 \langle \mathbf{u}, \mathbf{u} \rangle + \underbrace{k\langle \mathbf{u}, \mathbf{v} \rangle + k\langle \mathbf{v}, \mathbf{u} \rangle}_{=2k\langle \mathbf{u}, \mathbf{v} \rangle \text{ because } \langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle} + \langle \mathbf{v}, \mathbf{v} \rangle \\ &= k^2 \|\mathbf{u}\|^2 + 2k\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2 \left[\text{remember } \langle \mathbf{u}, \mathbf{u} \rangle = \|\mathbf{u}\|^2 \right]\end{aligned}$$

By Definition (4.1) Axiom (iv), we have $\langle \mathbf{w}, \mathbf{w} \rangle \geq 0$, which means that the inner product between the same vectors is ≥ 0 :

$$\langle k\mathbf{u} + \mathbf{v}, k\mathbf{u} + \mathbf{v} \rangle \geq 0 \quad [\text{remember } k\mathbf{u} + \mathbf{v} \text{ is a vector}]$$

Substituting this into the above expansion we have

$$k^2 \|\mathbf{u}\|^2 + 2k\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2 \geq 0 \quad (*)$$

We consider this as a quadratic equation by taking k as our variable. We can write this as

$$ak^2 + bk + c \geq 0 \text{ where } a = \|\mathbf{u}\|^2, b = 2\langle \mathbf{u}, \mathbf{v} \rangle \text{ and } c = \|\mathbf{v}\|^2$$

The quadratic is ≥ 0 , therefore we can substitute these $a = \|\mathbf{u}\|^2$, $b = 2\langle \mathbf{u}, \mathbf{v} \rangle$ and $c = \|\mathbf{v}\|^2$ into $(\dagger)$, which is given above by; $b^2 - 4ac \leq 0$:

$$\begin{aligned}b^2 - 4ac &= (2\langle \mathbf{u}, \mathbf{v} \rangle)^2 - 4\|\mathbf{u}\|^2\|\mathbf{v}\|^2 \leq 0 \\ (2\langle \mathbf{u}, \mathbf{v} \rangle)^2 &\leq 4\|\mathbf{u}\|^2\|\mathbf{v}\|^2 \\ 4(\langle \mathbf{u}, \mathbf{v} \rangle)^2 &\leq 4\|\mathbf{u}\|^2\|\mathbf{v}\|^2 \\ (\langle \mathbf{u}, \mathbf{v} \rangle)^2 &\leq \|\mathbf{u}\|^2\|\mathbf{v}\|^2 \quad [\text{cancelling 4's}] \\ \sqrt{(\langle \mathbf{u}, \mathbf{v} \rangle)^2} &\leq \sqrt{\|\mathbf{u}\|^2\|\mathbf{v}\|^2} \quad [\text{square root}] \\ |\langle \mathbf{u}, \mathbf{v} \rangle| &\leq \|\mathbf{u}\|\|\mathbf{v}\| \quad [\text{remember, } \sqrt{x^2} = |x|]\end{aligned}$$

This is our required result, the Cauchy–Schwarz inequality. ■

An important inequality for *addition* of norms of vectors is the following:

The Minkowski (pronounced ‘Minkofski’) or triangular inequality (4.7). Let V be an inner product space. For *all* vectors $\mathbf{u}$ and $\mathbf{v}$ we have

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$$



What does this mean?

In $\mathbb{R}^2$ we have (Fig. 4.5):

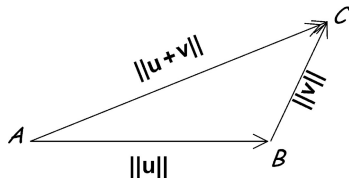


Figure 4.5

[Remember, this is the donkey theorem: if a donkey is at station A and the food is located at station C then the donkey would travel the shortest distance along AC not AB and then BC .]

In the above Example 4.6 (see Fig. 4.3(b)) we had the inequality:

$$\begin{array}{ccccc} \|\mathbf{u} + \mathbf{v}\| & \stackrel{=}{=} & 7.2111 & \leq & 7.2361 & \stackrel{=}{=} & \|\mathbf{u}\| + \|\mathbf{v}\| \\ & \text{by part (iv)} & & & & \text{by part (iii)} & \end{array}$$

Proof – Exercises 4.2. ■

Note that the Cauchy–Schwarz inequality establishes an inequality connecting the *multiplication* of vectors:

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

while the Minkowski inequality is an inequality relating *addition* of vectors:

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$$

4.2.2 Orthogonal vectors



Can you remember what orthogonal vectors meant in the Euclidean space $\mathbb{R}^n$?

Two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are said to be **orthogonal** or **perpendicular** $\Leftrightarrow$

$$(2.5) \quad \mathbf{u} \cdot \mathbf{v} = 0$$

Hence for the general vector space V with an inner product we have:

Definition (4.8). Two vectors $\mathbf{u}$ and $\mathbf{v}$ in the vector space V are said to be **orthogonal** $\Leftrightarrow \langle \mathbf{u}, \mathbf{v} \rangle = 0$

This is a fundamental and very useful result in linear algebra.

If vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, we say that $\mathbf{u}$ is orthogonal to $\mathbf{v}$, or vice versa that is $\mathbf{v}$ is orthogonal to $\mathbf{u}$ because $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle = 0$.

Consider the vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and $\mathbf{x}$ in $\mathbb{R}^2$ (Fig. 4.6):

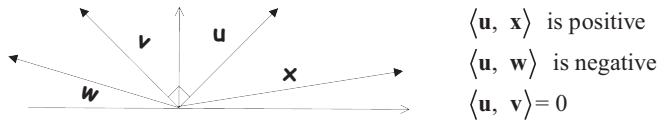


Figure 4.6

You may recall from chapter 2 that vectors acting in the same direction have a positive dot product and vectors in opposite directions have a negative dot product. If they are perpendicular ($\mathbf{u}$ and $\mathbf{v}$) then the dot product is zero.

Applications such as signal processing, communication and radar systems rely on inner products such as $S = \left\langle \frac{\mathbf{x}}{\|\mathbf{x}\|}, \frac{\mathbf{y}}{\|\mathbf{y}\|} \right\rangle$ where $\mathbf{x}$ and $\mathbf{y}$ are non-zero signals (vectors) in an inner product space. In the exercises, you are asked to show that this expression S lies between 0 and 1. S measures the degree to which the two signals are alike. A value of S close to 1 means that the signals are similar in nature. A value of S close to 0 means that the signals are very different but not necessarily orthogonal.

This expression S is useful in a matched filter detector. The detector uses S to compare a set of signals against a target signal to decide which signal is most like the target signal.

Another application of orthogonality is in information retrieval, because vector spaces can be used to represent documents. A document and a query term can be represented by vectors $\mathbf{d}$ and $\mathbf{q}$ respectively. If $\langle \mathbf{q}, \mathbf{d} \rangle = 0$ then vectors $\mathbf{q}$ and $\mathbf{d}$ are orthogonal, which means that the query term does not exist in the document.

To know that two functions or a set of functions are orthogonal is of incredibly powerful mathematical value, because you can use the theorems of linear algebra to develop approximations that are as close as you like to difficult functions that may be impossible to calculate in any other way. This is particularly true in the fields of mathematical physics such as Fourier analysis, polynomial series approximations and Legendre polynomials.

Orthogonality signifies a certain kind of ‘independence’, or a complete absence of interference. If the inner product is zero then one vector has *no* interference on the other.

Example 4.7

Show that the vectors $\mathbf{p} = x^2 + x + 1$ and $\mathbf{q} = -x^2 - x + 2$ are orthogonal in the vector space of polynomials of degree ≤ 2 , P_2 , where an inner product is given by

$$\langle \mathbf{p}, \mathbf{q} \rangle = c_2 d_2 + c_1 d_1 + c_0 d_0 \quad [\text{multiplying coefficients}]$$

$$(\mathbf{p} = c_2 x^2 + c_1 x + c_0 \text{ and } \mathbf{q} = d_2 x^2 + d_1 x + d_0).$$

Solution

For $\mathbf{p} = x^2 + x + 1$ and $\mathbf{q} = -x^2 - x + 2$ we have

$$\begin{aligned} \langle \mathbf{p}, \mathbf{q} \rangle &= \langle x^2 + x + 1, -x^2 - x + 2 \rangle \\ &= \langle (1)x^2 + (1)x + 1, (-1)x^2 + (-1)x + 2 \rangle \\ &= 1(-1) + 1(-1) + 1(2) = 0 \end{aligned}$$

(continued...)

We conclude that the given vectors $\mathbf{p}$ and $\mathbf{q}$ are orthogonal because $\langle \mathbf{p}, \mathbf{q} \rangle = 0$. We can illustrate this as shown in Fig. 4.7.

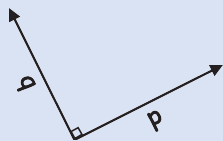


Figure 4.7

Example 4.8

Let the vector space of matrices of size 2 by 2, M_{22} , have an inner product defined by

$$\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{B}^T \mathbf{A}) \text{ where } \text{tr} \text{ is trace}$$

Show that the matrices $\mathbf{A} = \begin{pmatrix} -1 & -1 \\ 1 & 5 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}$ are orthogonal with respect to each other.

Remember, the trace of a matrix is the addition of leading diagonal entries.

Solution

$$\begin{aligned} \langle \mathbf{A}, \mathbf{B} \rangle &= \text{tr}(\mathbf{B}^T \mathbf{A}) = \text{tr} \left[\begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}^T \begin{pmatrix} -1 & -1 \\ 1 & 5 \end{pmatrix} \right] \\ &\stackrel{\text{transposing}}{=} \text{tr} \left[\begin{pmatrix} 1 & 3 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} -1 & -1 \\ 1 & 5 \end{pmatrix} \right] \stackrel{\text{multiplying}}{=} \text{tr} \begin{pmatrix} 2 & * \\ * & -2 \end{pmatrix} = 2 - 2 = 0 \end{aligned}$$

Hence matrices $\mathbf{A}$ and $\mathbf{B}$ are orthogonal, because $\langle \mathbf{A}, \mathbf{B} \rangle = 0$.

The structure of Euclidean vector spaces is exactly the same structure we find in other vector spaces. In the next example you will see this structure extended to accommodate the integrals of sine and cosine products.

Example 4.9

(Fourier series). Show that the vectors $f(t) = \cos(t)$ and $g(t) = \sin(t)$ are orthogonal in the set of continuous functions $C[0, \pi]$ with respect to the inner product given by

$$\langle \mathbf{f}, \mathbf{g} \rangle = \int_0^\pi f(t) g(t) dt$$

Solution

We have

$$\langle \mathbf{f}, \mathbf{g} \rangle = \langle \cos(t), \sin(t) \rangle = \int_0^\pi \cos(t) \sin(t) dt \quad (*)$$

How do we integrate $\cos(t) \sin(t)$?

Use the following trigonometric identity:

$$\cos(t) \sin(t) = \frac{1}{2} \sin(2t) \quad (\dagger)$$

Substituting this into the above we have

$$\begin{aligned} \int_0^\pi \cos(t) \sin(t) dt &= \frac{1}{2} \int_0^\pi \sin(2t) dt \quad [\text{applying the identity } (\dagger)] \\ &= \frac{1}{2} \left[-\frac{\cos(2t)}{2} \right]_0^\pi \left[\text{because } \int \sin(kt) dt = -\frac{\cos(kt)}{k} \right] \\ &= -\frac{1}{4} [\cos(2\pi) - \cos(0)] \quad \left[\begin{array}{l} \text{substituting limits and} \\ \text{taking out } -1/2 \end{array} \right] \\ &= -\frac{1}{4} [1 - 1] = 0 \quad [\text{because } \cos(2\pi) = \cos(0) = 1] \end{aligned}$$

By (*) we conclude that $\langle \mathbf{f}, \mathbf{g} \rangle = 0$, which means that $f(t) = \cos(t)$ and $g(t) = \sin(t)$ are orthogonal in $C[0, \pi]$. Hence sine and cosine functions are orthogonal in $C[0, \pi]$. This is an important result in the study of Fourier series.

Next we prove Pythagoras' Theorem which is applicable to vectors in general vector space with an inner product.

Pythagoras' Theorem (4.9). Let the vectors $\mathbf{u}$ and $\mathbf{v}$ be *orthogonal* in an inner product space V then

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

Proof.

Expanding the left hand side we have

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle \quad [\text{by definition of norm, } \|\mathbf{w}\|^2 = \langle \mathbf{w}, \mathbf{w} \rangle] \\ &= \langle \mathbf{u}, \mathbf{u} \rangle + \underbrace{\langle \mathbf{u}, \mathbf{v} \rangle}_{=0} + \underbrace{\langle \mathbf{v}, \mathbf{u} \rangle}_{=0} + \langle \mathbf{v}, \mathbf{v} \rangle \quad [\mathbf{u}, \mathbf{v} \text{ are orthogonal so } \langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle = 0] \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \quad [\text{remember, } \langle \mathbf{w}, \mathbf{w} \rangle = \|\mathbf{w}\|^2] \end{aligned}$$



Proposition (4.10). Every vector in an inner product space V is orthogonal to the zero vector, $\mathbf{0}$.

Proof.

Let $\mathbf{v}$ be an arbitrary vector in V . Then

$$\langle \mathbf{v}, \mathbf{0} \rangle = 0 \text{ [because by (4.2) part (i) we have } \langle \mathbf{v}, \mathbf{0} \rangle = 0]$$

Since $\mathbf{v}$ was arbitrary, every vector in V is orthogonal to the zero vector, $\mathbf{0}$. ■

4.2.3 Normalizing vectors

 *What is a unit vector?*

A unit vector $\mathbf{u}$ is a vector of length 1 or a norm of 1, that is $\|\mathbf{u}\| = 1$.

The process of converting a given vector into a unit vector is called **normalizing**.

Proposition (4.11). Every non-zero vector $\mathbf{w}$ in an inner product space V can be normalized by setting $\mathbf{u} = \frac{\mathbf{w}}{\|\mathbf{w}\|}$.

Proof – Exercises 4.2. ■

We write the normalized vector $\mathbf{w}$ as $\widehat{\mathbf{w}}$, which is pronounced as ‘w hat’. We have $\mathbf{u} = \widehat{\mathbf{w}}$.

Example 4.10

Normalize the vector $\mathbf{p} = 5x^2 - 2x + 1$ in P_2 with respect to the usual inner product of multiplying coefficients.

Solution

The normalized vector $\widehat{\mathbf{p}}$ is given by $\widehat{\mathbf{p}} = \mathbf{p} / \|\mathbf{p}\|$, where $\mathbf{p} = 5x^2 - 2x + 1$.

What is $\|\mathbf{p}\|$ equal to?

By the definition of the norm (4.3) we have $\|\mathbf{p}\|^2 = \langle \mathbf{p}, \mathbf{p} \rangle$, which we can evaluate by using the given inner product:

$$\begin{aligned} \|\mathbf{p}\|^2 &= \langle \mathbf{p}, \mathbf{p} \rangle = \langle 5x^2 - 2x + 1, 5x^2 - 2x + 1 \rangle \\ &= (5 \times 5) + (-2 \times (-2)) + (1 \times 1) = 30 \quad \text{[multiplying coefficients]} \end{aligned}$$

Taking the square root of $\|\mathbf{p}\|^2 = 30$ gives $\|\mathbf{p}\| = \sqrt{30}$. Hence the normalized (unit) vector is

$$\widehat{\mathbf{p}} = \frac{\mathbf{p}}{\|\mathbf{p}\|} = \frac{1}{\|\mathbf{p}\|} \mathbf{p} = \frac{1}{\sqrt{30}} \underbrace{(5x^2 - 2x + 1)}_{=\mathbf{p}}$$

4.2.4 Orthonormal set

A set of vectors which are orthogonal (perpendicular) to each other is called an **orthogonal set**.

A set of vectors in which *all* the vectors have a norm or length of 1 is called a **normalized set**.

A set of perpendicular unit vectors is called an **orthonormal set**. This is a set of vectors which are *both* orthogonal and normalized.

For example, the set of vectors $\mathbf{e}_1 = \begin{pmatrix} 1 & 0 \end{pmatrix}^T$ and $\mathbf{e}_2 = \begin{pmatrix} 0 & 1 \end{pmatrix}^T$ in $\mathbb{R}^2$ are orthonormal (perpendicular unit) vectors with the inner product as the dot product:

$$\langle \mathbf{e}_1, \mathbf{e}_2 \rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = (1 \times 0) + (0 \times 1) = 0$$

The vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ are orthogonal (perpendicular). The norms or lengths of these vectors:

$$\|\mathbf{e}_1\| = \|\mathbf{e}_2\| = 1$$

Thus the vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ are orthonormal (perpendicular unit vectors) because they are *both* orthogonal and normalized vectors.

Examples of orthonormal sets are shown in Fig. 4.8 for $\mathbb{R}^2$ and $\mathbb{R}^3$:

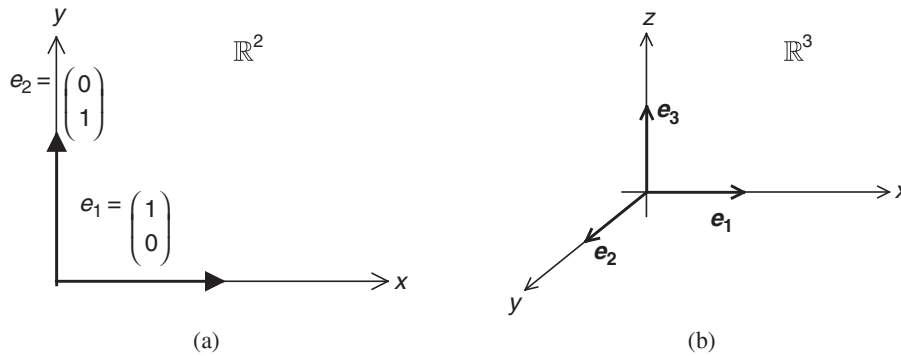


Figure 4.8

Remember that these are the standard basis vectors and they are also an orthonormal (perpendicular unit) basis for $\mathbb{R}^2$ and $\mathbb{R}^3$. These perpendicular unit vectors make a convenient basis as we will discuss in the next section.

Definition (4.12). Let V be a finite-dimensional vector space with an inner product. A set of vectors

$$B = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_n\}$$

for V is called an **orthonormal set** if they are

- (i) orthogonal, that is $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$ for $i \neq j$
- (ii) normalized, that is $\|\mathbf{u}_j\| = 1$ for $j = 1, 2, 3, \dots, n$

Normalized vectors have a norm (or length) of 1.

Orthogonal (different) vectors are orthogonal (perpendicular) to each other.

Example 4.11

Show that $B = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ forms an orthonormal (perpendicular unit) set of vectors for $\mathbb{R}^3$ with the inner product given by the dot product. (See Fig. 4.8(b).)

Solution

Remember, $\mathbf{e}_1 = (1 \ 0 \ 0)^T$, $\mathbf{e}_2 = (0 \ 1 \ 0)^T$ and $\mathbf{e}_3 = (0 \ 0 \ 1)^T$. We first check for orthogonality:

$$\langle \mathbf{e}_1, \mathbf{e}_2 \rangle = \mathbf{e}_1 \cdot \mathbf{e}_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = (1 \times 0) + (0 \times 1) + (0 \times 0) = 0$$

Similarly $\langle \mathbf{e}_1, \mathbf{e}_3 \rangle = \langle \mathbf{e}_2, \mathbf{e}_3 \rangle = 0$.

We have $\langle \mathbf{e}_1, \mathbf{e}_2 \rangle = \langle \mathbf{e}_1, \mathbf{e}_3 \rangle = \langle \mathbf{e}_2, \mathbf{e}_3 \rangle = 0$, therefore the set $B = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is orthogonal, which means that each of the vectors $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is perpendicular to the others.

What else do we need to prove?

The norm (or length) of each vector is 1, that is we need to show

$$\|\mathbf{e}_1\| = \|\mathbf{e}_2\| = \|\mathbf{e}_3\| = 1$$

Verify that the length of the vectors $\|\mathbf{e}_1\| = \|\mathbf{e}_2\| = \|\mathbf{e}_3\| = 1$.

Thus $B = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ forms an orthonormal (perpendicular unit) set of vectors in $\mathbb{R}^3$, because each vector is orthogonal to the others and all the vectors have a norm of 1.

This $B = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is *not* only an orthonormal set in $\mathbb{R}^3$, but actually forms an orthonormal *basis* for $\mathbb{R}^3$. The most useful type of basis is an orthonormal basis.

An orthonormal basis turns out to be a great tool used in the fields of differential equations and quantum mechanics.

Let's examine an orthonormal set for the vector space of polynomials.

Example 4.12

Let P_3 be the inner product space of polynomials of degree 3 or less with the usual inner product generated by multiplying coefficients. Let $\{\mathbf{p}, \mathbf{q}\}$ be an orthonormal set of vectors in P_3 . Find $\|\mathbf{p} + \mathbf{q}\|$.

Solution

Since $\{\mathbf{p}, \mathbf{q}\}$ is an orthonormal set, by definition the vectors are orthogonal and normalized (Fig. 4.9).

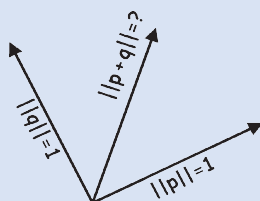


Figure 4.9

Applying Pythagoras (4.9), we have

$$\|p + q\|^2 = \|p\|^2 + \|q\|^2 = 1 + 1 = 2$$

Taking the square root gives $\|p + q\| = \sqrt{2}$.

The next example is more demanding because it relies on integration, and there are an infinite number of vectors – we have an infinite dimensional vector space. You will study this vector space in a subject called *Fourier series*. In Fourier series, a periodic function (a function which repeats itself in a given period) can be written as a linear combination of sines and cosines.

In the context of periodic functions, the inner product can be used to generate the Fourier coefficients.

You might be able to evaluate most of the integrals, but since this is not a course on calculus, the integrals are calculated for you.

Example 4.13

Let $C[0, 2\pi]$ be the vector space of continuous functions on the interval $[0, 2\pi]$ and the inner product on this vector space be defined by

$$\langle f_n, g_n \rangle = \int_0^{2\pi} f_n(t) g_n(t) dt$$

Let S be the set $\{1, \cos(t), \cos(2t), \cos(3t), \dots, \sin(t), \sin(2t), \sin(3t), \dots\}$. By letting $f_0 = \cos(0) = 1$, $f_n = \cos(nt)$ and $g_n = \sin(nt)$ show that the set S is orthogonal and normalize these vectors to create an orthonormal set.

[Results: for any positive integers n and m we have

- (i) $\int_0^{2\pi} \cos(nt) \sin(mt) dt = 0$,
- (ii) $\int_0^{2\pi} \cos(nt) dt = 0$,
- (iii) $\int_0^{2\pi} \sin(nt) dt = 0$,
- (iv) $\int_0^{2\pi} \cos(nt) \cos(mt) dt = 0$ provided $n \neq m$,
- (v) $\int_0^{2\pi} \sin(nt) \sin(mt) dt = 0$ provided $n \neq m$ and
- (vi) $\int_0^{2\pi} \cos^2(nt) dt = \int_0^{2\pi} \sin^2(nt) dt = \pi$.]

(continued...)

Solution

How do we show that the given set of vectors in S are orthogonal?

By using the integration results in the brackets above. First we test $\mathbf{f}_0$ with the other vectors:

$$\langle \mathbf{f}_0, \mathbf{f}_n \rangle = \int_0^{2\pi} [1 \times \cos(nt)] dt = \int_0^{2\pi} \cos(nt) dt \quad \stackrel{\text{by result (ii)}}{=} 0$$

$$\langle \mathbf{f}_0, \mathbf{g}_n \rangle = \int_0^{2\pi} [1 \times \sin(nt)] dt = \int_0^{2\pi} \sin(nt) dt \quad \stackrel{\text{by result (iii)}}{=} 0$$

Next we test $\mathbf{f}_n$ with $\mathbf{g}_m$:

$$\langle \mathbf{f}_n, \mathbf{g}_m \rangle = \int_0^{2\pi} \cos(nt) \sin(mt) dt \quad \stackrel{\text{by result (i)}}{=} 0$$

Lastly we test $\mathbf{f}_n$ with $\mathbf{f}_m$ and $\mathbf{g}_n$ with $\mathbf{g}_m$ where $m \neq n$:

$$\langle \mathbf{f}_n, \mathbf{f}_m \rangle = \int_0^{2\pi} \cos(nt) \cos(mt) dt \quad \stackrel{\text{by result (iv)}}{=} 0 \quad \text{provided } m \neq n$$

$$\langle \mathbf{g}_n, \mathbf{g}_m \rangle = \int_0^{2\pi} \sin(nt) \sin(mt) dt \quad \stackrel{\text{by result (v)}}{=} 0 \quad \text{provided } m \neq n$$

Hence the inner product of *all distinct* vectors is zero, which means that the set $\{1, \cos(t), \cos(2t), \cos(3t), \dots, \sin(t), \sin(2t), \sin(3t), \dots\}$ is orthogonal.

Normalizing:

First we determine $\|\mathbf{f}_0\|^2$, $\|\mathbf{f}_n\|^2$ and $\|\mathbf{g}_n\|^2$ and then we take the square root:

$$\begin{aligned} \|\mathbf{f}_0\|^2 &= \langle \mathbf{f}_0, \mathbf{f}_0 \rangle \\ &= \int_0^{2\pi} (1 \times 1) dt \\ &\stackrel{\text{integrating}}{=} [t]_0^{2\pi} = 2\pi - 0 = 2\pi \end{aligned}$$

$$\begin{aligned} \|\mathbf{f}_n\|^2 &= \langle \mathbf{f}_n, \mathbf{f}_n \rangle \\ &= \int_0^{2\pi} \cos(nt) \times \cos(nt) dt = \int_0^{2\pi} \cos^2(nt) dt \quad \stackrel{\text{by result (vi)}}{=} \pi \end{aligned}$$

$$\begin{aligned} \|\mathbf{g}_n\|^2 &= \langle \mathbf{g}_n, \mathbf{g}_n \rangle \\ &= \int_0^{2\pi} \sin(nt) \times \sin(nt) dt = \int_0^{2\pi} \sin^2(nt) dt \quad \stackrel{\text{by result (vi)}}{=} \pi \end{aligned}$$

Taking the square root of these results $\|\mathbf{f}_0\|^2 = 2\pi$, $\|\mathbf{f}_n\|^2 = \pi$ and $\|\mathbf{g}_n\|^2 = \pi$ gives $\|\mathbf{f}_0\| = \sqrt{2\pi}$, $\|\mathbf{f}_n\| = \sqrt{\pi}$ and $\|\mathbf{g}_n\| = \sqrt{\pi}$ respectively.

How can we normalize these (convert into unit vectors) $\mathbf{f}_0$, $\mathbf{f}_n$ and $\mathbf{g}_n$?

By using Proposition (4.11) $\mathbf{u} = \frac{\mathbf{w}}{\|\mathbf{w}\|}$ which means that we divide each vector by its norm. For all $n \geq 1$ we have

$$\frac{\mathbf{f}_0}{\|\mathbf{f}_0\|} = \frac{1}{\sqrt{2\pi}}, \quad \frac{\mathbf{f}_n}{\|\mathbf{f}_n\|} = \frac{\cos(nt)}{\sqrt{\pi}} \quad \text{and} \quad \frac{\mathbf{g}_n}{\|\mathbf{g}_n\|} = \frac{\sin(nt)}{\sqrt{\pi}}$$

Hence the set $\left\{ \frac{1}{\sqrt{2\pi}}, \frac{\cos(t)}{\sqrt{\pi}}, \frac{\cos(2t)}{\sqrt{\pi}}, \dots, \frac{\sin(t)}{\sqrt{\pi}}, \frac{\sin(2t)}{\sqrt{\pi}}, \dots \right\}$ is an orthonormal set of functions.

This orthonormal set is critical in the study of Fourier series. Note that the sines and cosines are orthogonal to each other in the vector space $C[0, 2\pi]$.

Summary

(4.8) Vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal $\Leftrightarrow \langle \mathbf{u}, \mathbf{v} \rangle = 0$.

(4.12) A set of perpendicular unit vectors

$$B = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_n\}$$

for vector space V is called an orthonormal set.



EXERCISES 4.2

(Brief solutions at end of book. Full solutions available at <http://www.oup.co.uk/companion/singh>.)

- Show that the following vectors in $\mathbb{R}^2$ are orthogonal with respect to the dot product. Also plot them on $\mathbb{R}^2$.
 - $\mathbf{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$
 - $\mathbf{u} = \begin{pmatrix} -2 \\ -3 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$
 - $\mathbf{u} = \begin{pmatrix} -4 \\ 5 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$
 - $\mathbf{u} = \begin{pmatrix} 2 \\ 7 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} -7 \\ 2 \end{pmatrix}$
- Show that the vectors $\mathbf{u} = \begin{pmatrix} a \\ b \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} -b \\ a \end{pmatrix}$ are orthogonal in $\mathbb{R}^2$ with respect to the dot product.
- Show that the following vectors in $\mathbb{R}^3$ form an orthogonal set with respect to the dot product.

$$\begin{aligned} \text{(a) } \mathbf{u} &= \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \mathbf{w} = \begin{pmatrix} 0 \\ 0 \\ 10 \end{pmatrix} \\ \text{(b) } \mathbf{u} &= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \quad \text{and} \quad \mathbf{w} = \begin{pmatrix} -3 \\ 10 \\ 7 \end{pmatrix} \end{aligned}$$

4. Let the vector space of size 2 by 2 matrices M_{22} have an inner product defined by $\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{B}^T \mathbf{A})$ where tr is the trace of the matrix.

(a) Show that the matrices $\mathbf{A} = \begin{pmatrix} 3 & 7 \\ 5 & 4 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 2 & 1 \\ 7 & -12 \end{pmatrix}$ are orthogonal.

(b) Determine $\|\mathbf{A} + \mathbf{B}\|$.

5. Determine the values of k so that the following vectors are orthogonal in $\mathbb{R}^4$ with respect to the dot product.

(a) $\mathbf{u} = (1 \ 2 \ 3 \ 4)^T$, $\mathbf{v} = (-2 \ 3 \ k \ 5)^T$ (b) $\mathbf{u} = (k \ -1 \ k \ 1)^T$, $\mathbf{v} = (2 \ 4 \ k \ 5)^T$

6. Let $C[0, 1]$ be the vector space of continuous functions on the interval $[0, 1]$ with an inner product defined by

$$\langle \mathbf{f}, \mathbf{g} \rangle = \int_0^1 f(x)g(x) \, dx$$

Verify the Cauchy–Schwarz inequality for

(a) $f(x) = x$ and $g(x) = x - 1$

(b) $f(x) = 1$ and $g(x) = e^x$

7. Show that any non-zero vectors $\mathbf{x}$ and $\mathbf{y}$ in an inner product space satisfy:

$$0 \leq \left\langle \frac{\mathbf{x}}{\|\mathbf{x}\|}, \frac{\mathbf{y}}{\|\mathbf{y}\|} \right\rangle \leq 1$$

8. Let $C[-\pi, \pi]$ be the vector space of continuous functions with an inner product defined by

$$\langle \mathbf{f}, \mathbf{g} \rangle = \int_{-\pi}^{\pi} f(x)g(x) \, dx$$

- (a) Show that $\mathbf{f}$ and $\mathbf{g}$ are orthogonal for $f(x) = \cos(x)$ and $g(x) = \sin(x)$.
 (b) Verify the Cauchy–Schwarz inequality for $\mathbf{f}$ and $\mathbf{g}$ given in part (a).
 (c) Verify the Minkowski (triangular) inequality for $\mathbf{f}$ and $\mathbf{g}$ given in part (a).
 (d) Normalize these vectors $\mathbf{f}$ and $\mathbf{g}$.

[You may use the following result: $\int_{-\pi}^{\pi} \sin^2(x) \, dx = \int_{-\pi}^{\pi} \cos^2(x) \, dx = \pi$]

9. Let $C[0, \pi]$ be the vector space of continuous functions on the interval $[0, \pi]$ with an inner product defined by

$$\langle \mathbf{f}, \mathbf{g}_n \rangle = \int_0^\pi f(t) g_n(t) dt$$

Let $f(t) = 1$ and $g_n(t) = \sin(2nt)$ where $n = 1, 2, 3, \dots$

Show that the set $\{1, \sin(2t), \sin(4t), \sin(6t), \dots\}$ is orthogonal. Normalize this set of vectors and write down the orthonormal set.

[You may use: $\int_0^\pi \sin(nt) \sin(mt) dt = 0$ provided $n \neq m$ and $\int_0^\pi \sin^2(nt) dt = \frac{\pi}{2}$]

10. Show that the following set S is an orthonormal set of 2 by 2 matrices M_{22} with an inner product defined by

$$\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{B}^T \mathbf{A})$$

$$S = \{\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}\} \quad \text{where} \quad \mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad \text{and} \\ \mathbf{D} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

11. Suppose a guitar string is modelled between 0 and 1 as shown in Fig. 4.10.

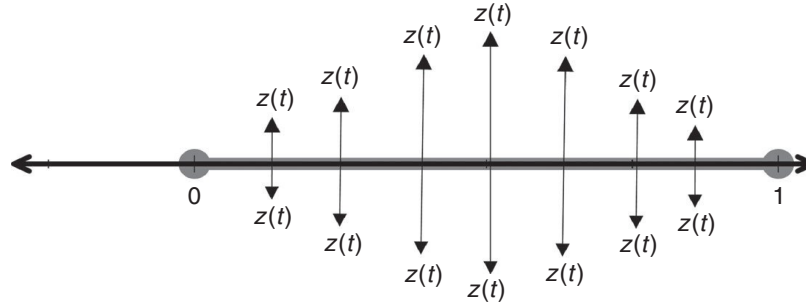


Figure 4.10

Consider a vertical force $f(t)$ at each point of the string and a vertical displacement of $z(t)$ at that point. The amount of energy released is given by the inner product:

$$\langle f(t), z(t) \rangle = \int_0^1 f(t) z(t) dt$$

Find the energy released if $f(t) = 100 \sin(100t)$ and $z(t) = \cos(10t)$.

[Hint: $2 \sin(A) \cos(B) = \sin(A+B) + \sin(A-B)$]

12. Prove Proposition (4.11).
13. Let $\mathbf{u}$ and $\mathbf{v}$ be orthogonal vectors in an inner product space. Prove that

$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

14. Let $\mathbf{u}$ and $\mathbf{v}$ be orthonormal vectors in an inner product space V . Determine
 (a) $\|\mathbf{u} + \mathbf{v}\|$ (b) $\|\mathbf{u} - \mathbf{v}\|$
15. (i) Let $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_n\}$ be an orthogonal set of vectors in an inner product space. Prove Pythagoras' theorem for these vectors, that is

$$\|\mathbf{u}_1 + \mathbf{u}_2 + \mathbf{u}_3 + \dots + \mathbf{u}_n\|^2 = \|\mathbf{u}_1\|^2 + \|\mathbf{u}_2\|^2 + \|\mathbf{u}_3\|^2 + \dots + \|\mathbf{u}_n\|^2$$

- (ii) Let $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \dots, \mathbf{f}_n\}$ be an orthonormal set of vectors in inner product space of $C[0, \pi]$. Find the length $\|\mathbf{f}_1 + \mathbf{f}_2 + \mathbf{f}_3 + \dots + \mathbf{f}_n\|$.
16. Prove that the vectors $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ are orthogonal $\Leftrightarrow \{k_1\mathbf{u}_1, k_2\mathbf{u}_2, \dots, k_n\mathbf{u}_n\}$ are orthogonal, where the k 's are non-zero scalars. Is the result still valid if any of k 's are zero?
17. Prove Minkowski inequality (4.7).

SECTION 4.3 Orthonormal Bases

By the end of this section you will be able to

- understand what is meant by an orthonormal basis
- find an orthonormal basis by using the Gram–Schmidt process

4.3.1 Introduction to an orthonormal bases

Why is an orthonormal basis important?

Generally, it is easier to work with an orthonormal basis (axes) rather than any other basis. For example, in $\mathbb{R}^2$, try working with a basis of $\mathbf{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 0.9 \\ 0.1 \end{pmatrix}$. Writing $\mathbf{w} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ in terms of these basis (axes) vectors $\mathbf{u}$ and $\mathbf{v}$ we have (Fig. 4.11).

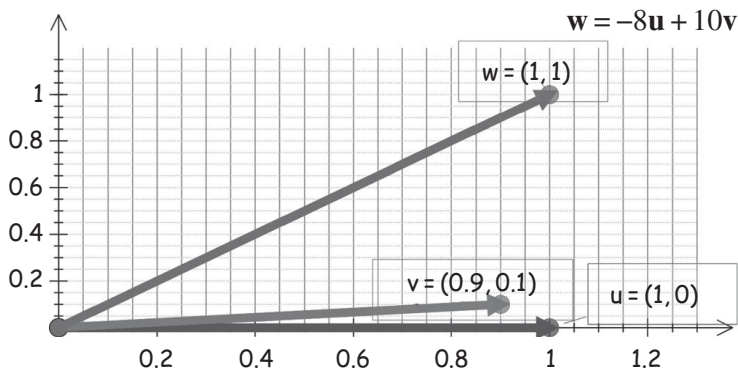


Figure 4.11

$$\begin{aligned}\mathbf{w} &= \begin{pmatrix} 1 \\ 1 \end{pmatrix} = -8 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 10 \begin{pmatrix} 0.9 \\ 0.1 \end{pmatrix} \\ &= -8\mathbf{u} + 10\mathbf{v}\end{aligned}$$

Writing $\mathbf{w} = -8\mathbf{u} + 10\mathbf{v}$ involves a lot more arithmetic than expressing this vector $\mathbf{w}$ in our usual orthonormal basis $\mathbf{e}_1$ and $\mathbf{e}_2$ as $\mathbf{w} = \mathbf{e}_1 + \mathbf{e}_2$ because $\mathbf{e}_1 = \begin{pmatrix} 1 & 0 \end{pmatrix}^T$ and $\mathbf{e}_2 = \begin{pmatrix} 0 & 1 \end{pmatrix}^T$ are the unit vectors in the x and y directions respectively.

In an n -dimensional vector space there are n orthogonal (perpendicular) axes or basis vectors. We will show that an orthogonal set of n vectors is automatically linearly independent, and therefore forms a legitimate basis (normalizing is just a matter of scale). Generally, it is easier to show that vectors are orthogonal rather than linearly independent.

For Fourier series (which is used in signal processing), an example of an orthogonal basis is

$$\{1, \sin(nx), \cos(nx)\} \text{ where } n \text{ is a positive integer}$$

? What do you think the term *orthonormal basis* means?

An orthonormal basis is a set of vectors which are normalized *and* are orthogonal to each other. They form a basis (axes) for the vector space.

Examples of orthonormal (perpendicular unit) basis are shown in Fig. 4.12 for $\mathbb{R}^2$ and $\mathbb{R}^3$:

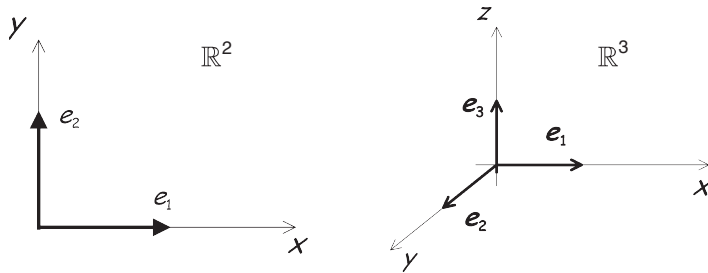


Figure 4.12

Note that our usual x , y and z axes are orthogonal to each other. The vectors $\{\mathbf{e}_1, \mathbf{e}_2\}$ form an orthonormal (perpendicular unit) basis for $\mathbb{R}^2$ and the set $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ forms an orthonormal (perpendicular unit) basis for $\mathbb{R}^3$.

In general, the set $B = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \dots, \mathbf{e}_n\}$ forms an orthonormal (perpendicular unit) basis for $\mathbb{R}^n$ with respect to the dot product. Remember, $\mathbf{e}_k = \begin{pmatrix} 0 & \dots & 1 & 0 & \dots \end{pmatrix}^T$ (1 in the k th position and zeros everywhere else.)

Definition (4.13). Let V be a finite dimensional vector space with an inner product. A set of basis vectors

$$B = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_n\}$$

for V is called an **orthonormal basis** if they are

- (i) Orthogonal, that is $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$ for $i \neq j$
- (ii) Normalized, that is $\|\mathbf{u}_j\| = 1$ for $j = 1, 2, 3, \dots, n$

 *What do you notice about this definition?*

It's the same as the definition of an orthonormal set (4.12) given in the last section, but this time the set of vectors are the *basis* vectors of the vector space.

 *Can you remember what the term *basis* means?*

It's a set of vectors which represent the axes of the vector space and satisfies two conditions:

- 1) Span the space
- 2) Linearly independent

Basis is like a coordinate system of a vector space – you can write any vector in terms of basis vectors.

For example, the normalized sine and cosine functions form an orthonormal basis for the vector space of periodic continuous functions, $C[0, 2\pi]$. (See Example 4.13 of section 4.2.4.)

To check that a given set of vectors form a basis for a vector space V can be tedious.

 *Why?*

Because normally we need to check two things about a basis:

- 1) the vectors span V
- 2) the vectors are linearly independent.

From the last chapter, we know that it is enough to show that for any n -dimensional vector space, n linearly *independent* vectors form a *basis* for this vector space. We only need to check condition (2). Next we show that an *orthogonal* set is linearly *independent*, that is

$$\text{orthogonality} \Rightarrow \text{linear independence}$$

4.3.2 Properties of an orthonormal basis

In this subsection, we prove that if we have an orthogonal set of vectors then they are linearly independent. We use this to prove that in an n -dimensional vector space, any set of n orthogonal non-zero vectors forms a basis for that vector space. (Generally, checking orthogonality is easier than checking linear independence.)

Proposition (4.14). If $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ is an *orthogonal* set of non-zero vectors in an inner product space then the elements of this set are *linearly independent*.

 *How do we prove this proposition?*

We want to prove that the vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ are linearly independent, so we consider the linear combination:

$$k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \dots + k_n\mathbf{v}_n = \mathbf{0}$$

and show that *all* the scalars are zero: $k_1 = 0, k_2 = 0, k_3 = 0, \dots$ and $k_n = 0$.



Why?

Because from chapter 3 we have:

(3.10). Vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are **linearly independent** $\Leftrightarrow$ the only solution to $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 + \dots + k_n\mathbf{v}_n = \mathbf{O}$ is $k_1 = k_2 = k_3 = \dots = k_n = 0$

Proof.

Consider the linear combination of the vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ and equate them to the zero vector, $\mathbf{O}$:

$$k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 + \dots + k_n\mathbf{v}_n = \mathbf{O} \quad (\dagger)$$

Consider the inner product of an arbitrary vector $\mathbf{v}_j$ in the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ with the zero vector given in $(\dagger)$.



What is the inner product $\langle k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 + \dots + k_n\mathbf{v}_n, \mathbf{v}_j \rangle$ equal to?

0, because

$$\langle k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 + \dots + k_n\mathbf{v}_n, \mathbf{v}_j \rangle = \langle \mathbf{O}, \mathbf{v}_j \rangle = 0 \quad (*)$$

We are given that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ is orthogonal, this means that the inner product between two *different* vectors is zero:

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0 \text{ if } i \neq j$$

Expanding $(*)$ and using $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ if $i \neq j$ we have

$$\begin{aligned} \langle k_1\mathbf{v}_1 + \dots + k_n\mathbf{v}_n, \mathbf{v}_j \rangle &= \langle k_1\mathbf{v}_1, \mathbf{v}_j \rangle + \dots + \langle k_j\mathbf{v}_j, \mathbf{v}_j \rangle + \dots + \langle k_n\mathbf{v}_n, \mathbf{v}_j \rangle \\ &\equiv k_1 \langle \mathbf{v}_1, \mathbf{v}_j \rangle + \dots + k_j \langle \mathbf{v}_j, \mathbf{v}_j \rangle + \dots + k_n \langle \mathbf{v}_n, \mathbf{v}_j \rangle \\ &\text{taking out scalars} \\ &= k_1 \underbrace{\langle \mathbf{v}_1, \mathbf{v}_j \rangle}_{=0} + \dots + k_j \underbrace{\langle \mathbf{v}_j, \mathbf{v}_j \rangle}_{=0} + \dots + k_n \underbrace{\langle \mathbf{v}_n, \mathbf{v}_j \rangle}_{=0} \\ &= k_1 (0) + 0 \dots 0 + k_j \|\mathbf{v}_j\|^2 + 0 \dots 0 + k_n (0) = k_j \|\mathbf{v}_j\|^2 \end{aligned}$$

The last line follows from the definition of the norm $\langle \mathbf{u}, \mathbf{u} \rangle = \|\mathbf{u}\|^2$. The *only* non-zero contribution is the inner product of the vectors $\mathbf{v}_j$ and $\mathbf{v}_j$. By $(*)$ we know that all this is equal to zero, which means that we have

$$k_j \|\mathbf{v}_j\|^2 = 0$$

$\mathbf{v}_j$ is one of the *non-zero* vectors among the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$, therefore $\|\mathbf{v}_j\|^2 \neq 0$ [not equal to 0]. Thus to satisfy the above $k_j \|\mathbf{v}_j\|^2 = 0$ we must have $k_j = 0$. Since $\mathbf{v}_j$ was an arbitrary vector, we conclude that *all* the scalars k 's in $(\dagger)$ must be zero, which proves that an *orthogonal* set of vectors is linearly independent. ■

? *What does this proposition mean?*

It means that *orthogonality* implies linear *independence*. For example, the vectors $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ are orthogonal (perpendicular), therefore linearly independent.

We can go further, as the next proposition states.

Corollary (4.15). In an n -dimensional inner product space V , any set of n orthogonal non-zero vectors forms a basis (or axes) for V .

? *Which tools do we use to prove this result?*

Theorem (3.22) (a) of the last chapter which says:

Any linearly *independent* set of n vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ forms a basis for V .

Proof.

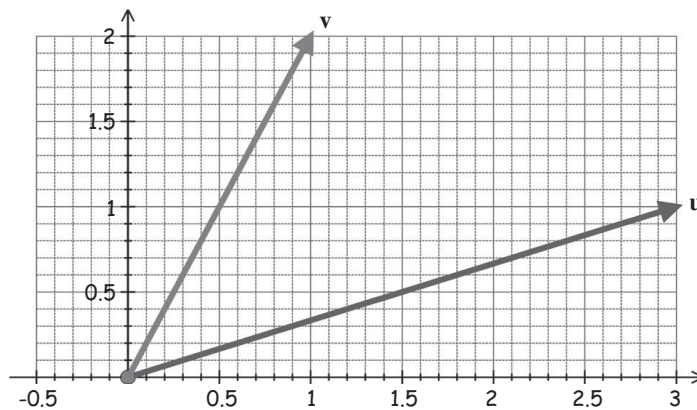
By the previous Proposition (4.14), we know that the set of n orthogonal vectors are linearly independent.

By Theorem (3.22) we conclude that n orthogonal vectors form a basis for V . ■

? *What does this corollary mean?*

If we have an n -dimensional vector space with an inner product then *any* n orthogonal (perpendicular) non-zero vectors form a set of basis (axes) vectors for that vector space.

However, vectors which are linearly independent may *not* be orthogonal. For example, the vectors $\mathbf{u} = (3 \ 1)^T$ and $\mathbf{v} = (1 \ 2)^T$ in $\mathbb{R}^2$ are linearly independent (we cannot make $\mathbf{v}$ from a scalar multiple of $\mathbf{u}$, and vice versa) but *not* orthogonal as you can see in Fig. 4.13.



These vectors are linearly independent but not orthogonal (not perpendicular).

Figure 4.13

Hence orthogonal vectors are independent, but independent vectors are *not* necessarily orthogonal. We have

$$\text{orthogonality} \Rightarrow \text{independence but independence} \not\Rightarrow \text{orthogonality}$$

Example 4.14

Show that $\mathbf{u} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$ are orthogonal in $\mathbb{R}^2$ with respect to the dot product and write down an orthonormal (perpendicular unit) basis (axes) for $\mathbb{R}^2$.

Solution

How do we show that the given vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal (perpendicular)?

By showing the inner (dot) product of these vectors is zero:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -1 \end{pmatrix} = (-1 \times (-1)) + (1 \times (-1)) = 0$$

Hence vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal (perpendicular).

How do we normalize these vectors?

By using Proposition (4.11) $\frac{\mathbf{w}}{\|\mathbf{w}\|}$, which means dividing a vector by its length.

What is length $\|\mathbf{u}\|$ equal to?

$$\|\mathbf{u}\|^2 = \langle \mathbf{u}, \mathbf{u} \rangle = \mathbf{u} \cdot \mathbf{u} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \end{pmatrix} = (-1)^2 + 1^2 = 2$$

Taking the square root gives $\|\mathbf{u}\| = \sqrt{2}$. Similarly we have $\|\mathbf{v}\| = \sqrt{2}$.

Thus the normalized (unit) vectors are

$$\frac{\mathbf{u}}{\|\mathbf{u}\|} = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix} \text{ and } \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

An orthonormal (perpendicular unit) basis (axes) for $\mathbb{R}^2$ is $\left\{ \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ -1 \end{pmatrix} \right\}$.

We can plot this orthonormal basis (vectors representing new axes) for $\mathbb{R}^2$ (Fig. 4.14).

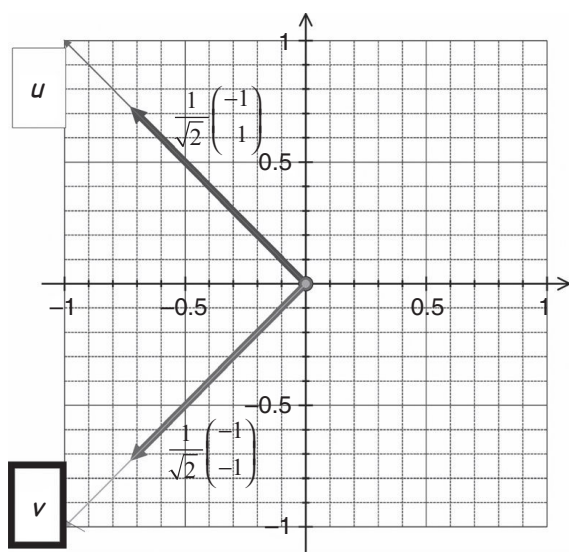


Figure 4.14

4.3.3 Generating orthogonal vectors

Consider two given vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^2$. We can project the vector $\mathbf{v}$ onto vector $\mathbf{u}$.



What does projection mean?

Projection is the procedure, for example, of showing a film on a screen; we say the film has been projected onto a screen. Similarly we project the vector $\mathbf{v}$ onto $\mathbf{u}$ as shown in Fig. 4.15.

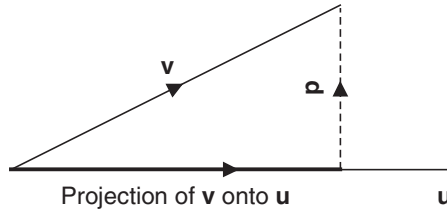


Figure 4.15



What is the projection of $\mathbf{v}$ onto $\mathbf{u}$ equal to?

Since it is in the direction of vector $\mathbf{u}$, it is a scalar multiple of $\mathbf{u}$. Let's nominate this scalar by the letter k , so we have

$$\text{Projection of } \mathbf{v} \text{ onto } \mathbf{u} = k\mathbf{u}$$

Let $\mathbf{p}$ be the perpendicular vector shown in Fig. 4.15. Adding the vectors gives

$$\mathbf{v} = k\mathbf{u} + \mathbf{p}$$

The vector $\mathbf{p}$ is orthogonal (perpendicular) to the vector $\mathbf{u}$, therefore $\langle \mathbf{u}, \mathbf{p} \rangle = 0$. This means that the projection of the perpendicular vector $\mathbf{p}$ onto vector $\mathbf{u}$ is zero.

Taking the inner product of this, $\mathbf{v} = k\mathbf{u} + \mathbf{p}$, with the vector $\mathbf{u}$, we have

$$\begin{aligned} \langle \mathbf{u}, \mathbf{v} \rangle &= \langle \mathbf{u}, (k\mathbf{u} + \mathbf{p}) \rangle \\ &= k\langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{p} \rangle \\ &= k\|\mathbf{u}\|^2 + 0 \quad \left[\begin{array}{l} \text{because } \mathbf{p} \text{ and } \mathbf{u} \text{ are perpendicular,} \\ \text{therefore } \langle \mathbf{u}, \mathbf{p} \rangle = 0 \text{ and } \langle \mathbf{u}, \mathbf{u} \rangle = \|\mathbf{u}\|^2 \end{array} \right] \\ &= k\|\mathbf{u}\|^2 \end{aligned}$$

Rearranging this $\langle \mathbf{u}, \mathbf{v} \rangle = k\|\mathbf{u}\|^2$ to make k the subject of the formula gives

$$k = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|^2} \quad (\dagger)$$

Rearranging the above $\mathbf{v} = k\mathbf{u} + \mathbf{p}$ to make $\mathbf{p}$ the subject:

$$\begin{aligned} \text{orthogonal vector } \mathbf{p} &= \mathbf{v} - k\mathbf{u} & [\mathbf{p} = \mathbf{v} - [\text{projection of } \mathbf{v} \text{ onto } \mathbf{u}]] \\ &= \mathbf{v} - \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|^2} \mathbf{u} & [\text{by } (\dagger)] \end{aligned}$$

We have perpendicular vector $\mathbf{p}$ (Fig. 4.16):

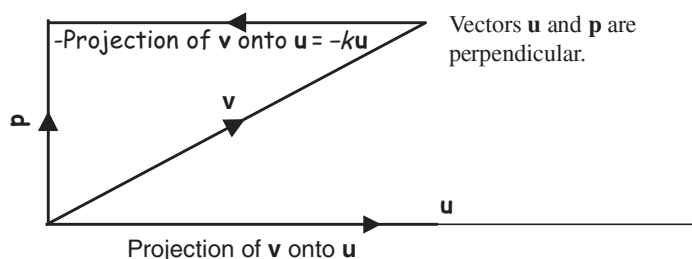


Figure 4.16

Hence we have created orthogonal vectors $\mathbf{p}$ and $\mathbf{u}$ out of the given non-orthogonal vectors $\mathbf{u}$ and $\mathbf{v}$.

It is important to note that the projection of orthogonal vectors is zero. This is what we were hinting at in the last section; *orthogonality signifies a certain kind of independence or a complete absence of interference.*

Example 4.15

Let $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ be vectors in $\mathbb{R}^2$. Construct a pair of orthogonal (perpendicular) vectors $\{\mathbf{p}_1, \mathbf{p}_2\}$ from this non-orthogonal set $\{\mathbf{v}_1, \mathbf{v}_2\}$.

Solution

We start with one of the given vectors, $\mathbf{v}_1$ say, and call this vector $\mathbf{p}_1$. Hence $\mathbf{p}_1 = \mathbf{v}_1$. We construct a vector which is orthogonal to $\mathbf{p}_1 = \mathbf{v}_1$.

By the above formula, orthogonal vector $\mathbf{p}_2 = \mathbf{v} - \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|^2} \mathbf{u}$ with $\mathbf{u} = \mathbf{v}_1 = \mathbf{p}_1$ and $\mathbf{v} = \mathbf{v}_2$:

$$\text{orthogonal vector } \mathbf{p}_2 = \mathbf{v}_2 - \frac{\langle \mathbf{p}_1, \mathbf{v}_2 \rangle}{\|\mathbf{p}_1\|^2} \mathbf{p}_1$$

Substituting $\mathbf{p}_1 = \mathbf{v}_1 = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ into this formula gives:

$$\text{orthogonal vector } \mathbf{p}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \frac{\left\langle \begin{pmatrix} 3 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\rangle}{\left\| \begin{pmatrix} 3 \\ 0 \end{pmatrix} \right\|^2} \begin{pmatrix} 3 \\ 0 \end{pmatrix} \quad (*)$$

Evaluating each component of (*):

$$\left\langle \begin{pmatrix} 3 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\rangle = \begin{pmatrix} 3 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} = (3 \times 1) + (0 \times 2) = 3, \quad \left\| \begin{pmatrix} 3 \\ 0 \end{pmatrix} \right\|^2 = \begin{pmatrix} 3 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 0 \end{pmatrix} = 3^2 + 0^2 = 9$$

Putting these values into (*) yields

$$\text{orthogonal vector } \mathbf{p}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \frac{3}{9} \begin{pmatrix} 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \frac{1}{3} \begin{pmatrix} 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 1-1 \\ 2-0 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

Illustrating the orthogonal vector $\mathbf{p}_2$ and the given vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ as shown in Fig. 4.17.

(continued...)

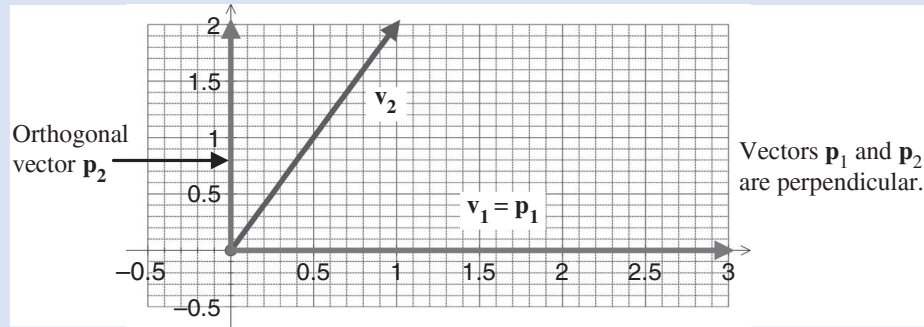


Figure 4.17

Note that we start with one of the given vectors, $\mathbf{v}_1$ say, then we create a vector orthogonal (perpendicular) to it by using the other given vector $\mathbf{v}_2$. We have achieved the following:

$$\left\{ \mathbf{v}_1 = \begin{pmatrix} 3 \\ 0 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\} \Rightarrow \left\{ \mathbf{p}_1 = \begin{pmatrix} 3 \\ 0 \end{pmatrix}, \mathbf{p}_2 = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \right\}$$

We can extend this procedure to any finite dimensional vector space and create an **orthogonal basis** for the vector space. Suppose we have a basis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$, and from this we want create an orthogonal basis $\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3, \dots, \mathbf{p}_n\}$. The procedure is:

1. Let $\mathbf{p}_1 = \mathbf{v}_1$, that is $\mathbf{p}_1$ equals one of the given vectors.
2. We create vector $\mathbf{p}_2$, which is orthogonal to $\mathbf{p}_1 = \mathbf{v}_1$, by using the second given vector $\mathbf{v}_2$. Hence we apply the above stated formula $\mathbf{p}_2 = \mathbf{v}_2 - \frac{\langle \mathbf{v}_2, \mathbf{p}_1 \rangle}{\|\mathbf{p}_1\|^2} \mathbf{p}_1$.
3. We create vector $\mathbf{p}_3$, which is orthogonal to both $\mathbf{p}_1 = \mathbf{v}_1$ and $\mathbf{p}_2$, by using the third given vector $\mathbf{v}_3$. The formula for this is similarly produced:

$$\mathbf{p}_3 = \mathbf{v}_3 - \frac{\langle \mathbf{v}_3, \mathbf{p}_1 \rangle}{\|\mathbf{p}_1\|^2} \mathbf{p}_1 - \frac{\langle \mathbf{v}_3, \mathbf{p}_2 \rangle}{\|\mathbf{p}_2\|^2} \mathbf{p}_2$$

4. We carry on producing vectors which are orthogonal (perpendicular) to the previous created vectors $\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_k$ by using the next given vector $\mathbf{v}_{k+1}$.

These steps are known as the Gram–Schmidt process.

4.3.4 The Gram–Schmidt process

Given any arbitrary basis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ for a finite dimensional inner product space, we can find an orthogonal basis $\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3, \dots, \mathbf{p}_n\}$ by the Gram–Schmidt process which is described next:

Gram–Schmidt process (4.16).

Let $\mathbf{p}_1 = \mathbf{v}_1$

$$\mathbf{p}_2 = \mathbf{v}_2 - \frac{\langle \mathbf{v}_2, \mathbf{p}_1 \rangle}{\|\mathbf{p}_1\|^2} \mathbf{p}_1$$

$$\mathbf{p}_3 = \mathbf{v}_3 - \frac{\langle \mathbf{v}_3, \mathbf{p}_1 \rangle}{\|\mathbf{p}_1\|^2} \mathbf{p}_1 - \frac{\langle \mathbf{v}_3, \mathbf{p}_2 \rangle}{\|\mathbf{p}_2\|^2} \mathbf{p}_2$$

$$\mathbf{p}_4 = \mathbf{v}_4 - \frac{\langle \mathbf{v}_4, \mathbf{p}_1 \rangle}{\|\mathbf{p}_1\|^2} \mathbf{p}_1 - \frac{\langle \mathbf{v}_4, \mathbf{p}_2 \rangle}{\|\mathbf{p}_2\|^2} \mathbf{p}_2 - \frac{\langle \mathbf{v}_4, \mathbf{p}_3 \rangle}{\|\mathbf{p}_3\|^2} \mathbf{p}_3$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots$$

$$\mathbf{p}_n = \mathbf{v}_n - \frac{\langle \mathbf{v}_n, \mathbf{p}_1 \rangle}{\|\mathbf{p}_1\|^2} \mathbf{p}_1 - \frac{\langle \mathbf{v}_n, \mathbf{p}_2 \rangle}{\|\mathbf{p}_2\|^2} \mathbf{p}_2 - \frac{\langle \mathbf{v}_n, \mathbf{p}_3 \rangle}{\|\mathbf{p}_3\|^2} \mathbf{p}_3 - \cdots - \frac{\langle \mathbf{v}_n, \mathbf{p}_{n-1} \rangle}{\|\mathbf{p}_{n-1}\|^2} \mathbf{p}_{n-1}$$

These vectors $\mathbf{p}$ are orthogonal, therefore a basis.



What else do we need to do to form an orthonormal basis?

Normalize $\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3, \dots, \mathbf{p}_n\}$ by using Proposition (4.11) which is $\frac{\mathbf{p}_j}{\|\mathbf{p}_j\|}$. [Dividing by their norms (lengths).]

The proof of Gram–Schmidt process (4.16) for an inner product space is given on the website <<http://www.oup.co.uk/companion/singh>>.

Example 4.16

Transform the basis $\mathbf{v}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ in $\mathbb{R}^2$ to an orthonormal (perpendicular unit) basis (axes) for $\mathbb{R}^2$ with respect to the dot product. Also plot this orthonormal basis and the given basis on two different graphs.

Solution

Using the above Gram–Schmidt process (4.16) means that we need to find vectors $\mathbf{p}_1$ and $\mathbf{p}_2$ which are orthogonal (perpendicular) to each other.

We have $\mathbf{p}_1 = \mathbf{v}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ which is one of the given vectors. By the above Gram–Schmidt process (4.16) we have

$$\mathbf{p}_2 = \mathbf{v}_2 - \frac{\langle \mathbf{v}_2, \mathbf{p}_1 \rangle}{\|\mathbf{p}_1\|^2} \mathbf{p}_1 \quad (*)$$

We need to evaluate each component of (*).
What is the inner product $\langle \mathbf{v}_2, \mathbf{p}_1 \rangle$ equal to?

$$\langle \mathbf{v}_2, \mathbf{p}_1 \rangle = \mathbf{v}_2 \cdot \mathbf{p}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix} = (1 \times 2) + (1 \times 1) = 3$$

(continued...)

What is $\|\mathbf{p}_1\|^2$ equal to?

$$\begin{aligned}\|\mathbf{p}_1\|^2 &= \langle \mathbf{p}_1, \mathbf{p}_1 \rangle = \mathbf{p}_1 \cdot \mathbf{p}_1 \\ &= \begin{pmatrix} 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix} = 2^2 + 1^2 = 5 \quad [\text{Pythagoras}]\end{aligned}$$

Substituting all these, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $\langle \mathbf{v}_2, \mathbf{p}_1 \rangle = 3$, $\|\mathbf{p}_1\|^2 = 5$ and $\mathbf{p}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, into (*) gives

$$\begin{aligned}\mathbf{p}_2 &= \mathbf{v}_2 - \frac{\langle \mathbf{v}_2, \mathbf{p}_1 \rangle}{\|\mathbf{p}_1\|^2} \mathbf{p}_1 \\ &= \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \frac{3}{5} \begin{pmatrix} 2 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 - 6/5 \\ 1 - 3/5 \end{pmatrix} = \begin{pmatrix} -1/5 \\ 2/5 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} -1 \\ 2 \end{pmatrix}\end{aligned}$$

Our vectors $\mathbf{p}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\mathbf{p}_2 = \frac{1}{5} \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ are an orthogonal (perpendicular) basis.

But how do we make these into an orthonormal (perpendicular unit) basis?

By normalizing these vectors (converting into unit vectors), which is achieved by dividing the vectors by their norms (lengths). Call these new unit vectors $\mathbf{u}_1$ and $\mathbf{u}_2$, and they are

$$\mathbf{u}_1 = \frac{\mathbf{p}_1}{\|\mathbf{p}_1\|} \quad \text{and} \quad \mathbf{u}_2 = \frac{\mathbf{p}_2}{\|\mathbf{p}_2\|}$$

From above $\|\mathbf{p}_1\|^2 = 5$ and taking square root gives $\|\mathbf{p}_1\| = \sqrt{5}$. We need to normalize the other perpendicular vector $\mathbf{p}_2$, which means we need to convert it to a unit vector:

$$\begin{aligned}\|\mathbf{p}_2\|^2 &= \langle \mathbf{p}_2, \mathbf{p}_2 \rangle = \frac{1}{5} \begin{pmatrix} -1 \\ 2 \end{pmatrix} \cdot \frac{1}{5} \begin{pmatrix} -1 \\ 2 \end{pmatrix} \\ &= \frac{1}{25} [(-1)^2 + 2^2] = \frac{5}{25} = \frac{1}{5}\end{aligned}$$

Thus $\|\mathbf{p}_2\| = \frac{1}{\sqrt{5}}$. Our normalized (unit) vectors are

$$\begin{aligned}\mathbf{u}_1 &= \frac{\mathbf{p}_1}{\|\mathbf{p}_1\|} = \frac{1}{\|\mathbf{p}_1\|} \mathbf{p}_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix} \\ \mathbf{u}_2 &= \frac{1}{\|\mathbf{p}_2\|} \mathbf{p}_2 = \frac{1}{\frac{1}{\sqrt{5}}} \frac{1}{5} \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} -1 \\ 2 \end{pmatrix} \quad \left[\text{By Law of Indices } \frac{1}{\sqrt{5}} 5 = \sqrt{5} \right]\end{aligned}$$

Our orthonormal (perpendicular unit) basis is the set $\{\mathbf{u}_1, \mathbf{u}_2\}$ where these are given above:

$$\mathbf{u}_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad \text{and} \quad \mathbf{u}_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

Plotting these orthonormal (perpendicular unit) basis and our given basis in $\mathbb{R}^2$ (Fig. 4.18):

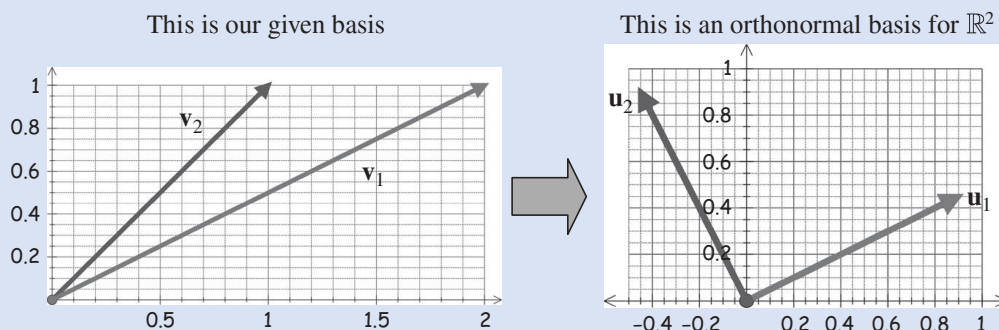


Figure 4.18

From Fig. 4.18 you should observe that $\mathbf{u}_1$ is in the same direction as the given vector $\mathbf{v}_1$ but has a length of 1. Remember, we start with $\mathbf{p}_1 = \mathbf{v}_1$, and $\mathbf{u}_1$ is a unit vector in that direction.

We can generally ignore the fraction (scalars) because vectors are orthogonal independent of scalars, as was established in question 16 of Exercises 4.2 which says:

Vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ are orthogonal $\Leftrightarrow \{k_1\mathbf{v}_1, k_2\mathbf{v}_2, \dots, k_n\mathbf{v}_n\}$ are orthogonal.

For example, in the above case we have $\mathbf{p}_2 = \frac{1}{5} \begin{pmatrix} -1 \\ 2 \end{pmatrix}$. Ignore the scalar (fraction) $\frac{1}{5}$ and normalize (convert to unit) the vector $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$:

$$\left\| \begin{pmatrix} -1 \\ 2 \end{pmatrix} \right\|^2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 2 \end{pmatrix} = (-1)^2 + 2^2 = 5 \text{ implies } \left\| \begin{pmatrix} -1 \\ 2 \end{pmatrix} \right\| = \sqrt{5}$$

Normalizing this vector gives $\mathbf{u}_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} -1 \\ 2 \end{pmatrix}$, which is the same unit vector $\mathbf{u}_2$ as in the above Example 4.16. It makes the arithmetic a lot easier if we ignore fractions or any other scalars.

The orthonormal basis is *not* unique. For instance, if you do *not* remove the scalar (fraction), you could have a *different* orthonormal basis. Also, if we apply the Gram-Schmidt process in reverse order, that is $\{\mathbf{v}_2, \mathbf{v}_1\}$, then the result may be a *different* orthonormal basis. In question 2 of Exercises 4.3, we have switched vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ of Example 4.16. See if you obtain the same or different set of vectors for the orthonormal basis.

We can also apply the Gram-Schmidt process to other vector spaces. Next we apply this to vector space of polynomials.

Example 4.17

Let P_2 be the vector space of polynomials of degree 2 or less with inner product given by:

$$\langle \mathbf{q}, \mathbf{r} \rangle = \int_{-1}^1 q(x) r(x) dx$$

Transform the standard basis $\{1, x, x^2\}$, which is *not* orthogonal, to an orthogonal basis.

Solution

Let $\mathbf{v}_1 = 1$, $\mathbf{v}_2 = x$, and $\mathbf{v}_3 = x^2$.

The first vector is given $\mathbf{p}_1 = \mathbf{v}_1 = 1$. Next we find a vector which is orthogonal to this by using (4.16) which is:

$$\mathbf{p}_2 = \mathbf{v}_2 - \frac{\langle \mathbf{v}_2, \mathbf{p}_1 \rangle}{\|\mathbf{p}_1\|^2} \mathbf{p}_1 \quad (*)$$

We need to find each of these components.

What is the inner product $\langle \mathbf{v}_2, \mathbf{p}_1 \rangle$ equal to?

$$\langle \mathbf{v}_2, \mathbf{p}_1 \rangle = \int_{-1}^1 x(1) dx = \int_{-1}^1 x dx = \left[\frac{x^2}{2} \right]_{-1}^1 = \frac{1}{2} [1^2 - (-1)^2] = 0$$

Since the inner product is zero, so the vectors $\mathbf{v}_2$ and $\mathbf{p}_1$ are already orthogonal. Putting this into (*) yields

$$\mathbf{p}_2 = \mathbf{v}_2 - (0) \mathbf{p}_1 = \mathbf{v}_2 = x \quad [\text{because we are given } \mathbf{v}_2 = x]$$

What else do we need to find?

The perpendicular vector $\mathbf{p}_3$ by applying Gram-Schmidt process (4.16) which is

$$\mathbf{p}_3 = \mathbf{v}_3 - \frac{\langle \mathbf{v}_3, \mathbf{p}_1 \rangle}{\|\mathbf{p}_1\|^2} \mathbf{p}_1 - \frac{\langle \mathbf{v}_3, \mathbf{p}_2 \rangle}{\|\mathbf{p}_2\|^2} \mathbf{p}_2 \quad (**)$$

Evaluating each of the components gives

$$\langle \mathbf{v}_3, \mathbf{p}_1 \rangle = \int_{-1}^1 x^2(1) dx = \int_{-1}^1 x^2 dx = \left[\frac{x^3}{3} \right]_{-1}^1 = \frac{1}{3} [1^3 - (-1)^3] = \frac{2}{3}$$

$$\langle \mathbf{v}_3, \mathbf{p}_2 \rangle = \int_{-1}^1 x^2(x) dx = \int_{-1}^1 x^3 dx = \left[\frac{x^4}{4} \right]_{-1}^1 = \frac{1}{4} [1^4 - (-1)^4] = 0$$

$$\|\mathbf{p}_1\|^2 = \langle \mathbf{p}_1, \mathbf{p}_1 \rangle = \int_{-1}^1 1(1) dx = [x]_{-1}^1 = [1 - (-1)] = 2$$

$$\|\mathbf{p}_2\|^2 = \langle \mathbf{p}_2, \mathbf{p}_2 \rangle = \int_{-1}^1 x(x) dx = \int_{-1}^1 x^2 dx = \frac{2}{3}$$

Substituting all these, $\langle \mathbf{v}_3, \mathbf{p}_1 \rangle = \frac{2}{3}$, $\|\mathbf{p}_1\|^2 = 2$, $\langle \mathbf{v}_3, \mathbf{p}_2 \rangle = 0$, and $\|\mathbf{p}_2\|^2 = \frac{2}{3}$ into (**) yields

$$\begin{aligned} \mathbf{p}_3 &= \mathbf{v}_3 - \frac{\langle \mathbf{v}_3, \mathbf{p}_1 \rangle}{\|\mathbf{p}_1\|^2} \mathbf{p}_1 - \frac{\langle \mathbf{v}_3, \mathbf{p}_2 \rangle}{\|\mathbf{p}_2\|^2} \mathbf{p}_2 \\ &= x^2 - \frac{2/3}{2} (1) - \frac{0}{2/3} x = x^2 - \frac{1}{3} \end{aligned}$$

Our three orthogonal polynomials (vectors) are

$$\mathbf{p}_1 = 1, \quad \mathbf{p}_2 = x \quad \text{and} \quad \mathbf{p}_3 = x^2 - \frac{1}{3}$$

In this example, we have transferred the standard basis to an orthogonal basis:

$$\{1, x, x^2\} \Rightarrow \{1, x, x^2 - 1/3\}$$

If we scale the above obtained orthogonal polynomials

$$\mathbf{p}_1 = 1, \quad \mathbf{p}_2 = x \quad \text{and} \quad \mathbf{p}_3 = x^2 - 1/3$$

so that at $x = 1$ they have a value of 1, we get

$$\mathbf{p}_1 = 1, \quad \mathbf{p}_2 = x \quad \text{and} \quad \mathbf{p}'_3 = \frac{1}{2}(3x^2 - 1)$$

These three orthogonal polynomials $\mathbf{p}_1$, $\mathbf{p}_2$ and $\mathbf{p}'_3$ are the first three **Legendre polynomials**. An important property of Legendre polynomials is that they must be orthogonal with respect to the inner product given in Example 4.17. Legendre polynomials crop up in differential equations.

There are also other orthogonal polynomials such as **Chebyshev** (pronounced 'chebishef') **polynomials** which are important in approximation theory. In approximation theory we try to provide the 'best' polynomial approximation $p(x)$ to a function $f(x)$ with a series expansion of orthogonal polynomials. By 'best' we mean that we construct a polynomial $p(x)$ so that the distance function or the error $\|f(x) - p(x)\|$ is as small as possible for all x in $[a, b]$, where $[a, b]$ is the interval our function is defined on.

This is different from a Taylor expansion, which gives the best approximation at a point.

Notice that the structure you see with dot product in Euclidean space is exactly the same structure you have with these polynomials.

It is interesting to know: if we have two given vector spaces U and V , are they essentially identical? We can think of U and V as identical if they have the same structure but only the nature of their elements or points differ. We say that vector spaces U and V are **isomorphic**.

For example, P_2 and $\mathbb{R}^3$ are isomorphic because they have the same structure but the elements such as $(1 \ 2 \ 3)^T$ in $\mathbb{R}^3$ differ from $1 + 2x + 3x^2$ in P_2 . We can plot these in three-dimensional space with our normal orthogonal x, y and z axes in $\mathbb{R}^3$ and constant, x and $x^2 - 1/3$ orthogonal axes in P_2 .

In the next chapter we will prove a very important result in linear algebra:

Every n -dimensional real vector space is isomorphic to $\mathbb{R}^n$.

This means that all the non-Euclidean vector spaces we have looked at in this chapter, such as the set of polynomials P_n , matrices M_{mn} and continuous functions on $C[a, b]$, are identical in structure to $\mathbb{R}^n$, provided they have the same dimension.

Proposition (4.17). Let $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ be an *orthonormal* set of vectors in an inner product space V of dimension n . Let $\mathbf{u}$ be a vector in V then

$$\mathbf{u} = \langle \mathbf{u}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{u}, \mathbf{v}_2 \rangle \mathbf{v}_2 + \langle \mathbf{u}, \mathbf{v}_3 \rangle \mathbf{v}_3 + \dots + \langle \mathbf{u}, \mathbf{v}_n \rangle \mathbf{v}_n$$

Proof – Exercises 4.3.

Note, the scalars are given by inner products. ■



Summary

(4.13). Let V be a finite dimensional vector space with an inner product. A set of basis vectors

$$B = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_n\}$$

for V is called an **orthonormal basis** if they are both orthogonal and normalized.

The Gram–Schmidt process (4.16) produces an orthogonal basis for a finite dimensional vector space.



EXERCISES 4.3

(Brief solutions at end of book. Full solutions available at <http://www.oup.co.uk/companion/singh>.)

1. Transform the following basis vectors in $\mathbb{R}^2$ to an orthonormal basis for $\mathbb{R}^2$ with respect to the dot product. Also plot the orthonormal basis.

(a) $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

(b) $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$

(c) $\mathbf{v}_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$

(d) $\mathbf{v}_1 = \begin{pmatrix} -2 \\ -5 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -3 \\ -1 \end{pmatrix}$

2. Consider Example 4.16 with the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ the other way around, that is

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ and } \mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

Transform these basis vectors in $\mathbb{R}^2$ to an orthonormal basis for $\mathbb{R}^2$ with respect to the dot product.

What do you notice about your answer in relation to Example 4.16?

3. Transform the basis $\mathbf{v}_1 = (1 \ 1 \ 1)^T$, $\mathbf{v}_2 = (1 \ 1 \ 0)^T$ and $\mathbf{v}_3 = (2 \ 0 \ 0)^T$ in $\mathbb{R}^3$ to an orthonormal (perpendicular unit) basis for $\mathbb{R}^3$ with respect to the dot product.
4. Transform the following basis vectors in $\mathbb{R}^3$ to an orthonormal basis for $\mathbb{R}^3$ with respect to the dot product.

(a) $\mathbf{v}_1 = (1 \ 0 \ 1)^T$, $\mathbf{v}_2 = (3 \ 1 \ 1)^T$ and $\mathbf{v}_3 = (-1 \ -1 \ -1)^T$

(b) $\mathbf{v}_1 = (2 \ 2 \ 2)^T$, $\mathbf{v}_2 = (-1 \ 0 \ -1)^T$ and $\mathbf{v}_3 = (-1 \ 2 \ 3)^T$

(c) $\mathbf{v}_1 = (1 \ 2 \ 0)^T$, $\mathbf{v}_2 = (2 \ 0 \ 2)^T$ and $\mathbf{v}_3 = (1 \ 0 \ 3)^T$

5. Transform the following vectors which span a subspace of $\mathbb{R}^4$ to an orthonormal basis for this subspace with respect to the dot product.
- (a) $\mathbf{v}_1 = (1 \ 0 \ 3 \ 0)^T$, $\mathbf{v}_2 = (1 \ 2 \ 1 \ 0)^T$ and $\mathbf{v}_3 = (2 \ 1 \ 0 \ 0)^T$
- (b) $\mathbf{v}_1 = (1 \ 1 \ 5 \ 2)^T$, $\mathbf{v}_2 = (-3 \ 3 \ 4 \ -2)^T$ and $\mathbf{v}_3 = (-1 \ -2 \ 2 \ 5)^T$
- (c) $\mathbf{v}_1 = (1 \ 2 \ 3 \ 4)^T$, $\mathbf{v}_2 = (2 \ 1 \ 1 \ 0)^T$ and $\mathbf{v}_3 = (3 \ 0 \ -1 \ 3)^T$
6. Normalize the orthogonal polynomials found in Example 4.17.
7. Transform the basis $\mathbf{v}_1 = x^2$, $\mathbf{v}_2 = x$, $\mathbf{v}_3 = 1$ to an orthogonal basis for P_2 with respect to the inner product given in Example 4.17. Compare this with the orthogonal basis obtained in Example 4.17.
8. Write the following polynomials in P_2 as linear combinations of the Legendre polynomials $\mathbf{p}_1 = 1$, $\mathbf{p}_2 = x$ and $\mathbf{p}_3 = \frac{1}{2}(3x^2 - 1)$:
- (a) $x^2 + x + 1$ (b) $2x^2 - 1$ (c) 3 (d) x^2 (e) $5x + 2$
9. Prove Proposition (4.17).

SECTION 4.4 Orthogonal Matrices

By the end of this section you will be able to

- understand what is meant by an orthogonal matrix
- factorize a given matrix

4.4.1 Orthogonal matrices

One of the most tedious problems in linear algebra is to find the inverse of a matrix. For a 3 by 3 or larger matrix, it becomes a monotonous task to determine the inverse by hand calculations. However, there is one set of matrices called *orthogonal matrices* where the inverse can be obtained by transposition of the matrix. It is straightforward to find the transpose of a matrix.

Orthogonal matrices arise naturally when working with orthonormal bases. Working with orthonormal bases is very handy because it allows you to use a formula like Pythagoras or it allows you to work in the field of Fourier series.

Orthogonal matrices are important in subjects such as numerical analysis because these matrices have good numerical stability.

Definition (4.18). A square matrix $\mathbf{Q} = (\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3 \ \cdots \ \mathbf{v}_n)$, whose columns $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n$ are orthonormal (perpendicular unit) vectors, is called an **orthogonal matrix**.

An example of an orthogonal matrix is the identity matrix.

Example 4.18

Let $\mathbf{Q} = (\mathbf{u}_1 \ \mathbf{u}_2 \ \mathbf{u}_3)$ where $\mathbf{u}_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, $\mathbf{u}_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$ and $\mathbf{u}_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$ which is an orthonormal basis for $\mathbb{R}^3$. Determine $\mathbf{Q}^T \mathbf{Q}$.

Solution

What does the notation $\mathbf{Q}^T$ mean?

The transpose of the matrix $\mathbf{Q}$. We have

$$\begin{aligned} \mathbf{Q}^T \mathbf{Q} &= (\mathbf{u}_1 \ \mathbf{u}_2 \ \mathbf{u}_3)^T (\mathbf{u}_1 \ \mathbf{u}_2 \ \mathbf{u}_3) \\ &= \begin{pmatrix} \mathbf{u}_1^T \\ \mathbf{u}_2^T \\ \mathbf{u}_3^T \end{pmatrix} (\mathbf{u}_1 \ \mathbf{u}_2 \ \mathbf{u}_3) \quad \left[\begin{array}{l} \text{When transposing the column} \\ \text{vector } \mathbf{u}_1 \text{ becomes the row} \\ \text{vector } \mathbf{u}_1^T. \end{array} \right] \\ &= \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{6} & -2/\sqrt{6} \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{pmatrix} \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{6} & 1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{6} & -1/\sqrt{2} \\ 1/\sqrt{3} & -2/\sqrt{6} & 0 \end{pmatrix} \end{aligned}$$

To evaluate this by hand is rather tedious, unless we can factor out some of the square roots:

$$\begin{aligned} \frac{1}{\sqrt{3}} &= \frac{\sqrt{2}}{\sqrt{2}\sqrt{3}} = \frac{\sqrt{2}}{\sqrt{6}} \quad \left[\begin{array}{l} \text{multiplying numerator} \\ \text{and denominator by } \sqrt{2} \end{array} \right] \\ \frac{1}{\sqrt{2}} &= \frac{\sqrt{3}}{\sqrt{3}\sqrt{2}} = \frac{\sqrt{3}}{\sqrt{6}} \quad \left[\begin{array}{l} \text{multiplying numerator} \\ \text{and denominator by } \sqrt{3} \end{array} \right] \end{aligned}$$

Replacing $1/\sqrt{3}$ and $1/\sqrt{2}$ in terms of $1/\sqrt{6}$ in the above:

$$\begin{aligned} \mathbf{Q}^T \mathbf{Q} &= \begin{pmatrix} \sqrt{2}/\sqrt{6} & \sqrt{2}/\sqrt{6} & \sqrt{2}/\sqrt{6} \\ 1/\sqrt{6} & 1/\sqrt{6} & -2/\sqrt{6} \\ \sqrt{3}/\sqrt{6} & -\sqrt{3}/\sqrt{6} & 0 \end{pmatrix} \begin{pmatrix} \sqrt{2}/\sqrt{6} & 1/\sqrt{6} & \sqrt{3}/\sqrt{6} \\ \sqrt{2}/\sqrt{6} & 1/\sqrt{6} & -\sqrt{3}/\sqrt{6} \\ \sqrt{2}/\sqrt{6} & -2/\sqrt{6} & 0 \end{pmatrix} \\ &= \frac{1}{\sqrt{6}} \begin{pmatrix} \sqrt{2} & \sqrt{2} & \sqrt{2} \\ 1 & 1 & -2 \\ \sqrt{3} & -\sqrt{3} & 0 \end{pmatrix} \frac{1}{\sqrt{6}} \begin{pmatrix} \sqrt{2} & 1 & \sqrt{3} \\ \sqrt{2} & 1 & -\sqrt{3} \\ \sqrt{2} & -2 & 0 \end{pmatrix} \quad \left[\begin{array}{l} \text{taking out a} \\ \text{scalar multiple} \\ 1/\sqrt{6} \end{array} \right] \\ &= \frac{1}{\sqrt{6}} \frac{1}{\sqrt{6}} \begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \end{aligned}$$



What do you notice about your final result in the above example?

We end up with the identity matrix $\mathbf{I}$, that is $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$. This is *no coincidence*. There is a general result which says that if matrix $\mathbf{Q}$ is orthogonal then $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$. We can also go the other way, that is if $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$ then matrix $\mathbf{Q}$ is orthogonal.

Proposition (4.19). Let $\mathbf{Q} = (\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3 \ \cdots \ \mathbf{v}_n)$ be a square matrix. Then $\mathbf{Q}$ is an orthogonal matrix $\Leftrightarrow \mathbf{Q}^T \mathbf{Q} = \mathbf{I}$.

 *How do we prove this result?*

We have $\Leftrightarrow$ in the statement, so we need to prove it both ways, $\Rightarrow$ and $\Leftarrow$.

Proof.

($\Rightarrow$). We assume that $\mathbf{Q} = (\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3 \ \cdots \ \mathbf{v}_n)$ is an orthogonal matrix, which means that $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is an orthonormal (perpendicular unit) set of vectors in $\mathbb{R}^n$. Required to prove $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$. We carry out the matrix multiplication:

$$\begin{aligned} \mathbf{Q}^T \mathbf{Q} &= (\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3 \ \cdots \ \mathbf{v}_n)^T (\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3 \ \cdots \ \mathbf{v}_n) \\ &= \begin{pmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_n^T \end{pmatrix} (\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3 \ \cdots \ \mathbf{v}_n) \quad \left[\begin{array}{l} \text{transposing to convert columns} \\ \text{to rows} \end{array} \right] \\ &= \begin{pmatrix} \mathbf{v}_1^T \mathbf{v}_1 & \mathbf{v}_1^T \mathbf{v}_2 & \cdots & \mathbf{v}_1^T \mathbf{v}_n \\ \mathbf{v}_2^T \mathbf{v}_1 & \mathbf{v}_2^T \mathbf{v}_2 & \cdots & \mathbf{v}_2^T \mathbf{v}_n \\ \vdots & \vdots & & \vdots \\ \mathbf{v}_n^T \mathbf{v}_1 & \mathbf{v}_n^T \mathbf{v}_2 & \cdots & \mathbf{v}_n^T \mathbf{v}_n \end{pmatrix} \quad \left[\begin{array}{l} \text{carrying out matrix} \\ \text{multiplication - row by} \\ \text{column} \end{array} \right] \end{aligned}$$

Remember, our destination is to show that the final matrix in the above is the identity.

 *What is the first entry $\mathbf{v}_1^T \mathbf{v}_1$ in the above matrix equal to?*

This is the dot product because we defined this in chapter 2:

$$(2.4) \quad \mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$$

Applying this to $\mathbf{v}_1^T \mathbf{v}_1$ we have

$$\mathbf{v}_1^T \mathbf{v}_1 = \mathbf{v}_1 \cdot \mathbf{v}_1$$

 *What is $\mathbf{v}_1 \cdot \mathbf{v}_1$ equal to?*

By the definition of the norm (length) of chapter 2: (2.8) $\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}}$

Hence $\mathbf{v}_1 \cdot \mathbf{v}_1 = \|\mathbf{v}_1\|^2$. Remember, $\mathbf{v}_1$ is an orthonormal (perpendicular unit) vector which means it has a length of 1. Therefore $\mathbf{v}_1 \cdot \mathbf{v}_1 = \|\mathbf{v}_1\|^2 = 1$. This means that the first entry of the matrix $\mathbf{Q}^T \mathbf{Q}$ is $\mathbf{v}_1^T \mathbf{v}_1 = 1$. Similarly all along the leading diagonal of $\mathbf{Q}^T \mathbf{Q}$, the entries are 1 because

$$\mathbf{v}_2^T \mathbf{v}_2 = \mathbf{v}_3^T \mathbf{v}_3 = \mathbf{v}_4^T \mathbf{v}_4 = \cdots = \mathbf{v}_n^T \mathbf{v}_n = 1$$

 *What does the second entry $\mathbf{v}_1^T \mathbf{v}_2$ in the matrix $\mathbf{Q}^T \mathbf{Q}$ equal to?*

Again, by (2.4) we have $\mathbf{v}_1^T \mathbf{v}_2 = \mathbf{v}_1 \cdot \mathbf{v}_2$.

? What is $\mathbf{v}_1 \cdot \mathbf{v}_2$ equal to?

Remember, we are assuming vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ are perpendicular unit vectors because they belong to an orthonormal set of vectors. Vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ are perpendicular so $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$. Similarly the dot product of two *different* vectors $\mathbf{v}_i$ and $\mathbf{v}_j$ is zero, that is

$$\mathbf{v}_i^T \mathbf{v}_j = \mathbf{v}_i \cdot \mathbf{v}_j = 0 \quad \text{provided } i \text{ does not equal } j$$

This means that *all* the remaining entries are equal to zero. Hence the matrix $\mathbf{Q}^T \mathbf{Q}$ has 1's along the leading diagonal and 0's everywhere else, which means that it is the identity matrix $\mathbf{I}$.

We have proved our required result $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$.

($\Leftarrow$). See Exercises 4.4. ■

? From this $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$, can you deduce any other result?

Since $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$, we conclude that the *inverse* of the matrix $\mathbf{Q}$ is $\mathbf{Q}^T$.

Proposition (4.20). $\mathbf{Q}$ is an orthogonal matrix $\Leftrightarrow \mathbf{Q}^{-1} = \mathbf{Q}^T$.

This is a powerful result because the inverse of any orthogonal matrix is the transpose of the matrix. It is a lot easier to transpose a matrix, rather than converting to reduced row echelon form.

Proof.

($\Rightarrow$). From the previous proposition, we have $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$ for an orthogonal matrix $\mathbf{Q}$. Remember, $\mathbf{Q}$ is a square matrix, so it has both left and right inverse and they are equal to each other.

By the definition of the inverse matrix of chapter 1:

(1.16). A square matrix $\mathbf{A}$ is said to be *invertible* if there is a matrix $\mathbf{B}$ such that $\mathbf{AB} = \mathbf{BA} = \mathbf{I}$ where $\mathbf{B}$ is denoted by $\mathbf{A}^{-1}$

Hence $\mathbf{Q}^T$ is the inverse of matrix $\mathbf{Q}$, which means that $\mathbf{Q}^{-1} = \mathbf{Q}^T$.

($\Leftarrow$). Required to prove that $\mathbf{Q}^{-1} = \mathbf{Q}^T$ implies that $\mathbf{Q}$ is orthogonal. Right multiplying $\mathbf{Q}^{-1} = \mathbf{Q}^T$ by $\mathbf{Q}$ gives $\mathbf{Q}^{-1} \mathbf{Q} = \mathbf{I} = \mathbf{Q}^T \mathbf{Q}$. By the above Proposition (4.19):

$$\mathbf{Q} \text{ is an orthogonal matrix } \Leftrightarrow \mathbf{Q}^T \mathbf{Q} = \mathbf{I}$$

We already have $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$, so $\mathbf{Q}$ is orthogonal. ■

4.4.2 Properties of orthogonal matrices

Example 4.19

Let $\mathbf{Q} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $\mathbf{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$. Determine the dot products:

- (i) $\mathbf{Q}\mathbf{u} \cdot \mathbf{Q}\mathbf{w}$ (ii) $\mathbf{u} \cdot \mathbf{w}$ (iii) What do you notice about your results?

Solution

(i) Evaluating the dot products:

$$\mathbf{Qu} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

$$\mathbf{Qw} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 4 \\ 2 \end{pmatrix}$$

$$\mathbf{QuQw} = \frac{1}{\sqrt{2}} \begin{pmatrix} 3 \\ -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \frac{1}{2} [(3 \times 4) + (-1 \times 2)] = 5$$

(ii) Similarly, we have

$$\mathbf{u} \cdot \mathbf{w} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 1 \end{pmatrix} = (1 \times 3) + (2 \times 1) = 5$$

(iii) Note that $\mathbf{Qu} \cdot \mathbf{Qw} = \mathbf{u} \cdot \mathbf{w}$.

This result $\mathbf{Qu} \cdot \mathbf{Qw} = \mathbf{u} \cdot \mathbf{w}$ for the last example is no coincidence. This is a general result that if $\mathbf{Q}$ is an orthogonal matrix then the dot product of $\mathbf{Qu}$ and $\mathbf{Qw}$ is exactly $\mathbf{u} \cdot \mathbf{w}$. Actually:

Proposition (4.21). Let $\mathbf{Q}$ be an n by n matrix and $\mathbf{u}$ and $\mathbf{v}$ be vectors in $\mathbb{R}^n$. Then

$$\mathbf{Q} \text{ is an orthogonal matrix} \Leftrightarrow \mathbf{Qu} \cdot \mathbf{Qw} = \mathbf{u} \cdot \mathbf{w}$$



What does this proposition mean?

This means that an orthogonal matrix preserves the dot product, and if the dot product is preserved then $\mathbf{Q}$ is an orthogonal matrix.

Proof.

($\Rightarrow$). Assume matrix $\mathbf{Q}$ is orthogonal. Required to prove $\mathbf{Qu} \cdot \mathbf{Qw} = \mathbf{u} \cdot \mathbf{w}$. Let us examine the left hand side of this $\mathbf{Qu} \cdot \mathbf{Qw} = \mathbf{u} \cdot \mathbf{w}$. We use

$$(2.4) \mathbf{u} \cdot \mathbf{w} = \mathbf{u}^T \mathbf{w}$$

Applying this formula (2.4) to $\mathbf{Qu} \cdot \mathbf{Qw}$ we have

$$\begin{aligned} \mathbf{Qu} \cdot \mathbf{Qw} &= (\mathbf{Qu})^T \mathbf{Qw} \\ &= (\mathbf{u}^T \mathbf{Q}^T) \mathbf{Qw} && \left[\text{by (1.19) (d) } (\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T \right] \\ &= \mathbf{u}^T (\mathbf{Q}^T \mathbf{Q}) \mathbf{w} && \left[\text{by (1.16)(a) } (\mathbf{AB}) \mathbf{C} = \mathbf{A} (\mathbf{BC}) \right] \\ &= \mathbf{u}^T (\mathbf{I}) \mathbf{w} && \left[\text{because } \mathbf{Q} \text{ is orthogonal so } \mathbf{Q}^T \mathbf{Q} = \mathbf{I} \right] \\ &= \mathbf{u}^T \mathbf{w} \stackrel{\text{By (2.4)}}{=} \mathbf{u} \cdot \mathbf{w} \end{aligned}$$

This proves that the dot product is preserved under matrix multiplication by an orthogonal matrix.

($\Leftarrow$). Now we go the other way. Assume $\mathbf{Qu} \cdot \mathbf{Qw} = \mathbf{u} \cdot \mathbf{w}$ and prove that $\mathbf{Q}$ is an orthogonal matrix. We have:

$$\begin{aligned}\mathbf{Qu} \cdot \mathbf{Qw} &= (\mathbf{Qu})^T \mathbf{Qw} \\ &= (\mathbf{u}^T \mathbf{Q}^T) \mathbf{Qw} \\ &= \mathbf{u}^T (\mathbf{Q}^T \mathbf{Q}) \mathbf{w} = \mathbf{u} \cdot \mathbf{w} \quad [\text{by assumption}]\end{aligned}$$

Using (2.4) $\mathbf{u} \cdot \mathbf{w} = \mathbf{u}^T \mathbf{w}$ on the right hand side of the last line, we have

$$\mathbf{u}^T (\mathbf{Q}^T \mathbf{Q}) \mathbf{w} = \mathbf{u}^T \mathbf{w} \text{ implies that } \mathbf{Q}^T \mathbf{Q} = \mathbf{I}$$

By the above Proposition (4.19): $\mathbf{Q}$ is an orthogonal matrix $\Leftrightarrow \mathbf{Q}^T \mathbf{Q} = \mathbf{I}$

We conclude that $\mathbf{Q}$ is an orthogonal matrix because $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$, which is our required result. ■

Example 4.20

Let $\mathbf{Q} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and $\mathbf{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Determine the lengths:

- (i) $\|\mathbf{u}\|$ (ii) $\|\mathbf{Qu}\|$ (iii) Plot the vectors $\mathbf{u}$ and $\mathbf{Qu}$ on $\mathbb{R}^2$.

Solution

(i) Using Pythagoras, we have

$$\|\mathbf{u}\|^2 = \mathbf{u} \cdot \mathbf{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 1^2 + 2^2 = 5 \quad \Rightarrow \quad \|\mathbf{u}\| = \sqrt{5}$$

(ii) Similarly we have

$$\|\mathbf{Qu}\|^2 = \mathbf{Qu} \cdot \mathbf{Qu} \quad \stackrel{\text{by Example 4.19}}{=} \quad \frac{1}{\sqrt{2}} \begin{pmatrix} 3 \\ -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 3 \\ -1 \end{pmatrix} = \frac{1}{2} [3^2 + (-1)^2] = 5$$

Hence $\|\mathbf{Qu}\| = \sqrt{5}$.

(iii) A plot of $\mathbf{u}$ and $\mathbf{Qu}$ is shown in Fig. 4.19.

Note that since lengths $\|\mathbf{u}\| = \|\mathbf{Qu}\| = \sqrt{5}$, the vectors $\mathbf{u}$ and $\mathbf{Qu}$ lie on the circle which has a radius of $\sqrt{5}$ from the origin.

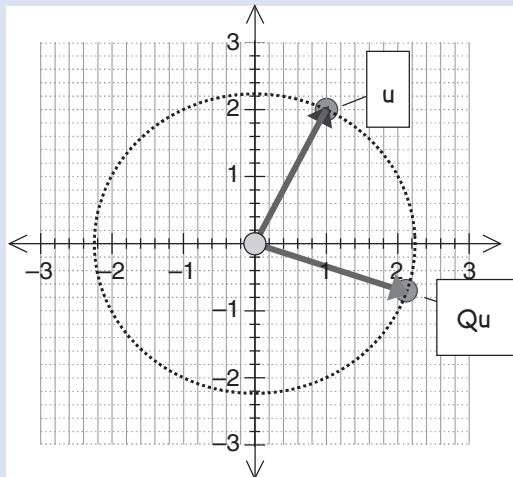


Figure 4.19

For an orthogonal matrix $\mathbf{Q}$ the length of vectors $\mathbf{Q}\mathbf{u}$ and $\mathbf{u}$ are equal.

Proposition (4.22). Let $\mathbf{Q}$ be an n by n matrix and $\mathbf{u}$ be a vector in $\mathbb{R}^n$. We have

$$\mathbf{Q} \text{ is an orthogonal matrix} \Leftrightarrow \|\mathbf{Q}\mathbf{u}\| = \|\mathbf{u}\|$$

? *What does this proposition mean?*

Applying an orthogonal matrix $\mathbf{Q}$ to a vector $\mathbf{u}$ may change the direction but *not* the length. An orthogonal matrix $\mathbf{Q}$ in $\mathbf{Q}\mathbf{u}$ will just rotate the vector $\mathbf{u}$ (Fig. 4.20).

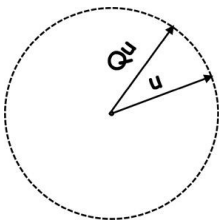


Figure 4.20

An orthogonal matrix preserves the length of a vector and if a transformation matrix preserves length then it is an orthogonal matrix.

Proof – Exercises 4.4.

■

4.4.3 Triangular matrices

? *What are triangular matrices?*

Definition (4.23). A triangular matrix is an n by n matrix where all entries to one side of the leading diagonal are zero.

For example, the following are triangular matrices:

$$(a) \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix} \text{ and } (b) \begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{pmatrix}$$

leading diagonal

(a) is an example of an upper triangular matrix.

(b) is an example of a lower triangular matrix.

Another type of matrix is a diagonal matrix.

? Do you know what is meant by a diagonal matrix?

Definition (4.24). A diagonal matrix is an n by n matrix where *all* entries to both sides of the leading diagonal are zero.

? Can you think of an example of a diagonal matrix?

The identity matrix $I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Another example is $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.

A diagonal matrix is both an upper and lower triangular matrix.

4.4.4 QR factorization

Let A be an m by n matrix with linearly independent columns. We can factorize matrix A into

$$(4.25) \quad A = QR$$

where Q is an orthogonal matrix and R is an upper triangular matrix.

? Why factorize the matrix A into an orthogonal matrix Q and upper triangular matrix R ?

1. So that we can efficiently solve linear systems such as $A\mathbf{x} = \mathbf{b}$. This linear system can be written as

$$(QR)\mathbf{x} = Q(R\mathbf{x}) = \mathbf{b} \quad (*)$$

Left multiplying this (*) by the inverse of Q which is $Q^{-1} = Q^T$ gives

$$\underbrace{Q^T Q}_{=I} (R\mathbf{x}) = R\mathbf{x} = Q^T \mathbf{b}$$

$Q^T \mathbf{b}$ is a vector $\mathbf{c}$ say, that is $Q^T \mathbf{b} = \mathbf{c}$. The given linear system $\mathbf{Ax} = \mathbf{b}$ has been reduced to solving $\mathbf{Rx} = \mathbf{c}$, where $\mathbf{R}$ is an upper triangular matrix. We can solve this system $\mathbf{Rx} = \mathbf{c}$ by back substitution because $\mathbf{R}$ is an upper triangular matrix:

$$\mathbf{Rx} = \mathbf{c} \text{ in matrix form is } \begin{pmatrix} r_{11} & r_{1n} \\ & \ddots & \\ & & r_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}$$

The advantage of this method is that once we have factorized matrix $\mathbf{A}$ then we can solve $\mathbf{Ax} = \mathbf{b}$ for various different vectors $\mathbf{b}$ in one go.

2. Factorizing a matrix is also applied in solving least squares approximation which is:

Find the vector $\mathbf{x}$ such that $\|\mathbf{Ax} - \mathbf{b}\|$ is a minimum

3. Additionally QR factorization is used to find the eigenvalues and eigenvectors (to be discussed in chapter 7) of a matrix by numerical means.

We need to find an orthogonal matrix $\mathbf{Q}$ and an upper triangular matrix $\mathbf{R}$ such that $\mathbf{A} = \mathbf{QR}$.

 *How can we find such matrices?*

Since $\mathbf{Q}$ is an orthogonal matrix, its columns are orthonormal (perpendicular unit) vectors:

$$\mathbf{Q} = (\mathbf{q}_1 \ \mathbf{q}_2 \ \mathbf{q}_3 \ \cdots \ \mathbf{q}_n)$$

 *But how do we get these orthonormal vectors from the matrix $\mathbf{A}$?*

Well, let matrix $\mathbf{A} = (\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3 \ \cdots \ \mathbf{a}_n)$ where the vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ are linearly independent. Then by applying the Gram–Schmidt process we can transform these $\mathbf{a}$ vectors into $\mathbf{q}$ vectors because Gram–Schmidt produces orthonormal vectors.

Applying the Gram–Schmidt process (4.16) and normalization from the last section we have:

Let $\mathbf{a}_1 = \frac{\mathbf{q}_1}{\|\mathbf{q}_1\|}$. Rearranging this gives $\mathbf{a}_1 = r_{11}\mathbf{q}_1$ where $r_{11} = \frac{1}{\|\mathbf{q}_1\|}$.

Also $\mathbf{q}_2 = \frac{1}{\|\mathbf{q}_2\|} \left[\mathbf{a}_2 - \frac{\langle \mathbf{v}_2, \mathbf{q}_1 \rangle}{\|\mathbf{q}_1\|^2} \mathbf{q}_1 \right]$. Transposing this gives

$$\mathbf{a}_2 = \frac{\langle \mathbf{v}_2, \mathbf{q}_1 \rangle}{\|\mathbf{q}_1\|^2} \mathbf{q}_1 + \|\mathbf{q}_2\| \mathbf{q}_2$$

We can write this as $\mathbf{a}_2 = r_{12}\mathbf{q}_1 + r_{22}\mathbf{q}_2$ where $r_{12} = \frac{\langle \mathbf{v}_2, \mathbf{q}_1 \rangle}{\|\mathbf{q}_1\|^2}$ and $r_{22} = \|\mathbf{q}_2\|$.

Expanding by applying the Gram–Schmidt process we have

$$\begin{aligned} \mathbf{a}_1 &= r_{11}\mathbf{q}_1 \\ \mathbf{a}_2 &= r_{12}\mathbf{q}_1 + r_{22}\mathbf{q}_2 \\ \mathbf{a}_3 &= r_{13}\mathbf{q}_1 + r_{23}\mathbf{q}_2 + r_{33}\mathbf{q}_3 \\ &\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\ \mathbf{a}_n &= r_{1n}\mathbf{q}_1 + r_{2n}\mathbf{q}_2 + r_{3n}\mathbf{q}_3 + \cdots + r_{nn}\mathbf{q}_n \end{aligned}$$

We can write this in matrix form as

$$\begin{aligned}
 \mathbf{A} &= (\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3 \ \cdots \ \mathbf{a}_n) \\
 &= (r_{11}\mathbf{q}_1 \quad r_{12}\mathbf{q}_1 + r_{22}\mathbf{q}_2 \quad r_{13}\mathbf{q}_1 + r_{23}\mathbf{q}_2 + r_{33}\mathbf{q}_3 \quad \cdots \quad r_{1n}\mathbf{q}_1 + r_{2n}\mathbf{q}_2 + r_{3n}\mathbf{q}_3 + \cdots + r_{nn}\mathbf{q}_n) \\
 &= (\underbrace{\mathbf{q}_1 \ \mathbf{q}_2 \ \mathbf{q}_3 \ \cdots \ \mathbf{q}_n}_{=\mathbf{Q}}) \underbrace{\begin{pmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ 0 & r_{22} & \cdots & r_{2n} \\ 0 & 0 & \ddots & \vdots \\ 0 & \vdots & \cdots & r_{nn} \end{pmatrix}}_{=\mathbf{R}} \\
 &= \mathbf{QR}
 \end{aligned}$$

The problem is how do we find the upper triangular matrix $\mathbf{R}$? Since $\mathbf{Q}$ is orthogonal we know the inverse of matrix $\mathbf{Q}$ is $\mathbf{Q}^T$. Left multiplying both sides of $\mathbf{A} = \mathbf{QR}$ by $\mathbf{Q}^T$ gives

$$\mathbf{Q}^T \mathbf{A} = \underbrace{\mathbf{Q}^T \mathbf{Q}}_{=\mathbf{I}} \mathbf{R} = \mathbf{I} \mathbf{R} = \mathbf{R}$$

Hence the upper triangular matrix $\mathbf{R}$ is given by

$$(4.26) \quad \mathbf{R} = \mathbf{Q}^T \mathbf{A}$$

Let us apply this $\mathbf{QR}$ factorization to a particular matrix $\mathbf{A}$.

Example 4.21

Find the $\mathbf{QR}$ factorization of $\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$.

Solution

The column vectors of $\mathbf{A}$ are $\mathbf{a}_1 = (1 \ 1 \ 1)^T$, $\mathbf{a}_2 = (1 \ 1 \ 0)^T$ and $\mathbf{a}_3 = (2 \ 0 \ 0)^T$. These are the same vectors given in question 3 of Exercises 4.3. By applying the Gram-Schmidt process, we found the following orthonormal vectors:

$$\mathbf{Q} = (\mathbf{q}_1 \ \mathbf{q}_2 \ \mathbf{q}_3) \text{ where } \mathbf{q}_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \mathbf{q}_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \text{ and } \mathbf{q}_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

The matrix $\mathbf{Q}$ and its transpose were simplified in Example 4.18:

$$\mathbf{Q} = \frac{1}{\sqrt{6}} \begin{pmatrix} \sqrt{2} & 1 & \sqrt{3} \\ \sqrt{2} & 1 & -\sqrt{3} \\ \sqrt{2} & -2 & 0 \end{pmatrix} \text{ and } \mathbf{Q}^T = \frac{1}{\sqrt{6}} \begin{pmatrix} \sqrt{2} & \sqrt{2} & \sqrt{2} \\ 1 & 1 & -2 \\ \sqrt{3} & -\sqrt{3} & 0 \end{pmatrix}$$

Why do we need to find the transpose of the matrix $\mathbf{Q}$?

Because the upper triangular matrix $\mathbf{R} = \mathbf{Q}^T \mathbf{A}$:

$$\begin{aligned}\mathbf{R} = \mathbf{Q}^T \mathbf{A} &= \frac{1}{\sqrt{6}} \begin{pmatrix} \sqrt{2} & \sqrt{2} & \sqrt{2} \\ 1 & 1 & -2 \\ \sqrt{3} & -\sqrt{3} & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \\ &= \frac{1}{\sqrt{6}} \begin{pmatrix} 3\sqrt{2} & 2\sqrt{2} & 2\sqrt{2} \\ 0 & 2 & 2 \\ 0 & 0 & 2\sqrt{3} \end{pmatrix}\end{aligned}$$

You may like to check the following by carrying out the matrix multiplication:

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} = \mathbf{QR} = \frac{1}{\sqrt{6}} \begin{pmatrix} \sqrt{2} & 1 & \sqrt{3} \\ \sqrt{2} & 1 & -\sqrt{3} \\ \sqrt{2} & -2 & 0 \end{pmatrix} \frac{1}{\sqrt{6}} \begin{pmatrix} 3\sqrt{2} & 2\sqrt{2} & 2\sqrt{2} \\ 0 & 2 & 2 \\ 0 & 0 & 2\sqrt{3} \end{pmatrix}$$

Summary

Orthogonal matrices are square matrices whose columns are orthonormal (perpendicular unit) vectors.

We can factorize an m by n matrix $\mathbf{A}$ whose column vectors are linearly independent into $\mathbf{A} = \mathbf{QR}$ where $\mathbf{Q}$ is an orthogonal matrix and $\mathbf{R}$ is an upper triangular matrix.

EXERCISES 4.4

(Brief solutions at end of book. Full solutions available at <http://www.oup.co.uk/companion/singh>.)

1. In each of the following cases, show that $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$:

(a) $\mathbf{Q} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

(b) $\mathbf{Q} = \frac{1}{5} \begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix}$

(c) $\mathbf{Q} = \begin{pmatrix} \cos(\theta) & \sin(\theta) \\ \sin(\theta) & -\cos(\theta) \end{pmatrix}$

2. Determine $\mathbf{Q}^{-1}$ for each of the matrices in question 1.

3. Determine which of the following matrices are orthogonal. If they are orthogonal find their inverse.

$$(a) \mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$(b) \mathbf{B} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & -1 \\ 1 & -1 \end{pmatrix}$$

$$(c) \mathbf{C} = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \\ -1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \\ 0 & 1/\sqrt{3} & -2/\sqrt{6} \end{pmatrix}$$

4. Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. What can you say about the size and angle between the column vectors of $\mathbf{A}$, where $\mathbf{A}$ is an orthogonal matrix?
5. Find the third column vector of an orthogonal matrix whose first two columns are:

$$\begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix} \text{ and } \begin{pmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{pmatrix}$$

6. Determine $\mathbf{Q}\mathbf{u}$ where $\mathbf{Q} = \begin{pmatrix} \cos(\theta) & \sin(\theta) \\ \sin(\theta) & -\cos(\theta) \end{pmatrix}$ and $\mathbf{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Plot your result on $\mathbb{R}^2$. What effect does the orthogonal matrix $\mathbf{Q}$ have on the vector $\mathbf{u}$?

7. Let $\mathbf{A} = \begin{pmatrix} 2 & -1 & -1 \\ 2 & 0 & 2 \\ 2 & -1 & 3 \end{pmatrix}$. The column vectors of $\mathbf{A}$ were the vectors given in Exercises 4.3 question 4(b).

- Determine the $\mathbf{QR}$ factorization of matrix $\mathbf{A}$.
 - Solve the linear system $\mathbf{Ax} = \mathbf{b}$ where $\mathbf{b} = (-3 \ 8 \ 9)^T$.
 - Solve the linear system $\mathbf{Ax} = \mathbf{b}$ where $\mathbf{b} = (-5 \ 12 \ 11)^T$.
8. Prove the converse of Proposition (4.19).
9. Prove Proposition (4.22).
10. Prove that if $\mathbf{A}$ is a square matrix with the property $\|\mathbf{Au}\| = 1$ for all unit vectors $\mathbf{u}$ then $\mathbf{A}$ is an orthogonal matrix.
11. If $\mathbf{Q}$ is orthogonal prove that $\mathbf{Q}^T$ is also orthogonal.
12. Prove that the columns and rows of an orthogonal matrix are linearly independent.



MISCELLANEOUS EXERCISES 4

(Brief solutions at end of book. Full solutions available at <http://www.oup.co.uk/companion/singh>.)

- 4.1. Decide whether the following statement is *True* or *False*. Justify your answer.

Every linearly independent set in $\mathbb{R}^n$ is an orthogonal set.

Washington State University, USA
(part question)

4.2. The vectors

$$\mathbf{X}_1 = [1, 1, 1, 1], \mathbf{X}_2 = [1, 0, 0, 1], \mathbf{X}_3 = [0, 2, 1, -1]$$

are a basis of a subspace U of $\mathbb{R}^4$. Find an orthogonal basis of U .

University of Ottawa, Canada

4.3. Let $S = \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \\ 2 \end{bmatrix} \right\}$.

- (a) Find a basis for S .
- (b) Find an orthonormal basis for S .

Purdue University, USA

4.4. (a) Which bases of a Euclidean space V are called orthogonal? Orthonormal?

(b) Show that $\mathbf{f}_1 = \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$, $\mathbf{f}_2 = \begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix}$ and $\mathbf{f}_3 = \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix}$ form a basis of $\mathbb{R}^3$.

- (c) Find the orthogonal basis of $\mathbb{R}^3$ which is the output of the Gram–Schmidt orthogonalization applied to the basis from part (b). (The inner product on $\mathbb{R}^3$ is the standard one.)

University of Dublin, Ireland

4.5. (a) Prove that an orthogonal set of non-zero vectors in $\mathbb{R}^n$ is linearly independent.

- (b) Let $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be an orthonormal basis of $\mathbb{R}^n$. Show that for $\mathbf{w} \in \mathbb{R}^n$,

$$\mathbf{w} = (\mathbf{v}_1 \cdot \mathbf{w}) \mathbf{v}_1 + (\mathbf{v}_2 \cdot \mathbf{w}) \mathbf{v}_2 + \dots + (\mathbf{v}_n \cdot \mathbf{w}) \mathbf{v}_n$$

University of Manchester, UK
(part question)

4.6. (a) Define what it means for vectors $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ in $\mathbb{R}^3$ to be orthonormal.

- (b) Apply the Gram–Schmidt process to the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 5 \\ 4 \\ 6 \end{bmatrix}$$

in $\mathbb{R}^3$ to find an orthonormal set in $\mathbb{R}^3$.

University of Southampton, UK
(part question)

4.7. Consider the real vector space $\mathbb{R}^4$ with real inner product given by

$$\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + x_2 y_2 + x_3 y_3 + x_4 y_4$$

for $\mathbf{x} = (x_1, x_2, x_3, x_4), \mathbf{y} = (y_1, y_2, y_3, y_4) \in \mathbb{R}^4$.

- Define the norm of a vector $\mathbf{x} = (x_1, x_2, x_3, x_4) \in \mathbb{R}^4$ with respect to the above inner product. What is the norm of $(2, -3, 5, 1)$?
- When do we say that two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^4$ are orthogonal? Are $(0, 1, -1, 1)$ and $(6, 3, 3, 27)$ orthogonal?
- What is an orthonormal set of vectors in $\mathbb{R}^4$?
Is $\left\{ (0, 0, 1, 0), \left(\frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}, 0, \frac{1}{\sqrt{6}} \right), \left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}, 0, 0 \right) \right\}$ an orthonormal set? Justify your answer.
- Use the Gram–Schmidt process to construct an orthonormal basis for $\mathbb{R}^4$ starting from the basis

$$\{(1, 0, 0, 0), (1, 1, 0, 1), (0, 1, 1, 1), (0, 1, -1, 0)\}$$

City University, London, UK

4.8. Consider the real vector space $M(2, 2)$ with real inner product given by

$$\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{B}^T \mathbf{A})$$

for all $\mathbf{A}, \mathbf{B} \in M(2, 2)$.

[$M(2, 2)$ is M_{22} which is the set of 2×2 matrices and tr is the trace of the matrix.]

- Define the norm of a matrix $\mathbf{A} \in M(2, 2)$ with respect to the above inner product. What is the norm of $\begin{pmatrix} 2 & -3 \\ 5 & 1 \end{pmatrix}$?
- When do we say that two matrices $\mathbf{A}, \mathbf{B} \in M(2, 2)$ are orthogonal (with respect to the above inner product)? Are $\begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}$ and $\begin{pmatrix} 6 & 3 \\ 3 & 27 \end{pmatrix}$ orthogonal?
- What is an orthonormal set of matrices in $M(2, 2)$ (with respect to the above inner product)? Is

$$\left\{ \mathbf{A}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \mathbf{A}_2 = \begin{pmatrix} 0 & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} \end{pmatrix}, \mathbf{A}_3 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \right\}$$

an orthonormal set? Justify your answer.

- Let $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Using (c) and the fact that $\{\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3, \mathbf{B}\}$ is a basis for $M(2, 2)$, find a matrix $\mathbf{A}_4$ such that $\{\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3, \mathbf{A}_4\}$ is an orthonormal basis for $M(2, 2)$.

(e) Find $\lambda_1, \lambda_2, \lambda_3, \lambda_4 \in \mathbb{R}$ such that

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \lambda_1 \mathbf{A}_1 + \lambda_2 \mathbf{A}_2 + \lambda_3 \mathbf{A}_3 + \lambda_4 \mathbf{A}_4$$

City University, London, UK

4.9. Suppose $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$ are orthonormal vectors in $\mathbb{R}^4$ (not $\mathbb{R}^3$).

- (a) What is the length of the vector $\mathbf{v} = 2\mathbf{q}_1 - 3\mathbf{q}_2 + 2\mathbf{q}_3$?
- (b) What four vectors does Gram–Schmidt produce when it orthonormalizes the vectors $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \mathbf{u}$?
- (c) If $\mathbf{u}$ in part (b) is the vector $\mathbf{v}$ in part (a), why does the Gram–Schmidt break down?

Massachusetts Institute of Technology, USA (part question)

4.10. Consider the vectors

$$\mathbf{v}_1 = [0, 1, 0, 1, 0], \mathbf{v}_2 = [0, 1, 1, 0, 0], \mathbf{v}_3 = [0, 1, 0, 1, 1]$$

in $\mathbb{R}^5$. Find $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ in $\mathbb{R}^5$ such that $\mathbf{w}_i \cdot \mathbf{w}_j = 0$ for $i \neq j$ (i and j between 1 and 3), and such that $\text{Span}(\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}) = \text{Span}(\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\})$ for $i = 1, 2, 3$.

University of California, Berkeley, USA

4.11. (a) Can you find vectors $\mathbf{v} = [v_1, v_2]$ and $\mathbf{w} = [w_1, w_2]$ in $\mathbb{R}^2$ which are orthogonal with respect to the inner product $\langle \mathbf{v}, \mathbf{w} \rangle = 3v_1w_1 + 2v_2w_2$, but not orthonormal with respect to the usual dot product?

(b) If your answer to (a) is ‘yes’, find $\mathbf{v}$ and $\mathbf{w}$. If your answer to (a) is ‘no’, give a reason why you cannot find $\mathbf{v}$ and $\mathbf{w}$.

University of Toronto, Canada

4.12. [Edited version of question.] Let P_3 be the set of all polynomials with degree less than or equal to 3. We will think of them as a subspace of $C[-1, 1]$, the continuous functions from the interval $[-1, 1]$ to the reals. P_3 has an inner product given by $\langle \mathbf{f}, \mathbf{g} \rangle = \int_{-1}^1 f(x)g(x)dx$. P_3 has what we call a *standard* basis $\{1, \mathbf{x}, \mathbf{x}^2, \mathbf{x}^3\}$. Let V be the space spanned by $\{1, \mathbf{x}\}$. Find an orthonormal basis for V .

Johns Hopkins University, USA

4.13. Define, for $\mathbf{A}$ and $\mathbf{B}$ in $M_{2,2}(\mathbb{R})$

$$\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{A}^T \mathbf{B})$$

(Recall, if $\mathbf{M} = [m_{ij}] \in M_{n,n}(F)$, $\text{tr}(\mathbf{M}) = \sum_{i=1}^n m_{ii}$.)

- (a) Show that $\langle \mathbf{A}, \mathbf{B} \rangle$ is an inner product on $M_{2,2}(\mathbb{R})$.
- (b) Find the distance between $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 3 & 3 \\ 1 & 2 \end{bmatrix}$ in this inner product space.

- (c) Find the angle between $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 3 & 3 \\ 1 & 2 \end{bmatrix}$ in this inner product space.

University of Toronto, Canada

- 4.14.** Consider the linear space V of polynomials $\mathbf{f}(t) = c_0 + c_1t + c_2t^2$ on $0 \leq t \leq 1$ with inner product

$$\langle \mathbf{f}, \mathbf{g} \rangle = \int_0^1 \mathbf{f}(t) \mathbf{g}(t) dt$$

Find a basis for the subspace S of all $\mathbf{f}$ in V orthogonal to $1 + t$ satisfying the additional restriction equation $\mathbf{f}(\frac{1}{2}) = 0$.

University of Utah, USA

- 4.15.** Define the inner product of two polynomials $\mathbf{f}$ and $\mathbf{g}$ by the rule

$$\langle \mathbf{f}, \mathbf{g} \rangle = \int_0^1 \mathbf{f}(x) \mathbf{g}(1-x) dx$$

Using this definition of the inner product, find an orthogonal basis for the vector space of all polynomials of degree ≤ 2 .

Columbia University, New York, USA

Sample questions

- 4.16.** Explain why $\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{AB})$ where $\mathbf{A}, \mathbf{B} \in M_{22}$ is *not* an inner product.
4.17. Prove Proposition (4.17).
4.18. Let P_3 be the vector space of cubic polynomials with the inner product given by

$$\langle \mathbf{f}, \mathbf{g} \rangle = \int_{-1}^1 f(x) g(x) dx$$

Convert the standard basis $\{1, x, x^2, x^3\}$ to an orthonormal basis for P_3 .
 [The resulting polynomials are called the **Legendre** (normalized) polynomials.]

- 4.19.** Find an orthogonal basis for the column space $C(\mathbf{A})$ of the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 3 & 3 \\ 2 & 5 & 1 & 0 \\ 0 & 4 & 0 & 0 \end{bmatrix}$$

- 4.20.** Let V be an inner product space. Prove that if the vector $\mathbf{u}$ is orthogonal for every $\mathbf{v} \in V$ then $\mathbf{u} = \mathbf{O}$.
4.21. Let V be an inner product space. Prove that if the vector $\mathbf{u}$ is orthogonal to $\mathbf{v}$ then every scalar multiple of $\mathbf{u}$ is also orthogonal to $\mathbf{v}$.

- 4.22.** Let $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ be a set of orthonormal vectors in an inner product space V . Prove that

$$\langle x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_n\mathbf{v}_n, y_1\mathbf{v}_1 + y_2\mathbf{v}_2 + \cdots + y_n\mathbf{v}_n \rangle = x_1y_1 + x_2y_2 + \cdots + x_ny_n$$

where x_i and y_i are scalars.

- 4.23.** The distance function between two vectors $\mathbf{u}$ and $\mathbf{v}$ in an inner product space V is denoted as $d(\mathbf{u}, \mathbf{v})$ and defined in (4.4) as

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$$

Prove the following results:

- (a) If $\mathbf{u} \neq \mathbf{v}$ then $d(\mathbf{u}, \mathbf{v}) > 0$.
- (b) $d(\mathbf{u}, \mathbf{v}) \leq d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$ where $\mathbf{w} \in V$.